Where Does Evolvability Reside? Organisms, Encounters, and Ecological History in Self-Reproducing Cellular Automata

Separating candidate production, stable transmission, and ecological performance in controlled encounters

Abstract

Encounters can generate new forms; useful heritable variation requires both transmission and ecological value. We investigate these stages in evoloops, a deterministic cellular automaton whose interacting reproductive structures can alter descendants. An initial two-organism atlas produced candidates in six of 40 contacting configurations but no confirmed stable novelty, including after delayed extraction and broader descendant search. A prospective three-identity atlas produced candidate words in 14/36 main and 4/18 held-out encounters, with none in 90 controls. Serial isolation confirmed pair-relative inherited novelty in two main configurations and none held out. One yielded a word absent from all three inputs; the other recovered a known third identity absent from its input pair. Main candidate rank reversals and inherited interaction contrasts did not transfer held out. A subsequent ecological calibration recovered the original identity from 63/68 selected scored cores and all five baseline controls, passing its frozen gate. All 20 paired ecological conditions completed. The known-word recovery exceeded both originating inputs under the frozen finite population-score criterion; the catalogue-novel identity did not. However, 36/40 scheduled clearings removed no occupied cells, none immediately reduced the counted population, and reserved confirmation conditions duplicated development trajectories under rotation. Independent ecological benefit confirmation therefore remains unresolved. The atlases also differ in inputs and measurement, preventing causal attribution of their yield difference. These results separate candidate production, recovered transmission, and a conditional performance advantage while bounding claims about partner-dependent evolvability.

1. Introduction

A system's capacity for evolutionary change depends on more than producing new forms. For adaptive evolution, differences must pass to descendants and improve performance in a relevant environment. Separating these steps is particularly important when reproduction itself depends on interactions: an encounter may create a novel structure, help it reproduce, or destroy it before its differences can persist.

Evoloops make these possibilities observable. They are self-reproducing structures embedded in a two-dimensional cellular automaton. A circulating signal sequence participates in constructing offspring, while contact between reproductive structures can change the sequence and shape of the resulting loops. All events follow fixed local rules (Sayama, 1999). The model therefore lets us intervene on encounters without inserting a separate mutation operator.

Where, then, does the capacity to generate heritable variation reside? An organism may carry a transferable bias in the variants it produces. That bias may be expressed differently against different partners. Previous ecological conditions may shape both the organisms present and the encounters they experience. These are compatible contributions, whose identification requires different comparisons. We organize them as three hypotheses:

H1, a contribution transferable with the organism under common encounter conditions; H2, a contribution specific to organism–partner combinations; and H3, historical effects mediated by changes in organisms and encounter distributions.

The present experiments address the encounter comparison. We first cross two input organisms across a defined spatial panel, then follow candidate variants through isolation and repeated reproduction. Crossing partners asks whether the outcome associated with changing one organism depends on the other. Following descendants asks whether any such pattern concerns temporary structural changes or transmitted variation. A calibration examines whether early extraction or restricted descendant search explains failed transmission. A fresh crossed atlas then uses three naturally generated, separately verified inherited identities, new configurations, and the calibrated descendant search to examine how far the original result depends on its input panel.

The completed studies recover both negative and positive transmission outcomes. The original candidates did not yield confirmed stable novelty under either calibrated search, whereas two configurations in the fresh atlas did. One produces a word absent from the input catalogue; the other recovers a known catalogue word absent from its input pair. Neither result transfers as an inherited-yield difference to the held-out configurations. A subsequent paired ecological comparison tests both inherited outcomes against both originating inputs. Its score calibration passes, and one outcome meets the frozen finite proxy criterion, but ineffective disturbances and rotationally duplicated confirmation conditions limit its interpretation. Together these studies establish what organism and history comparisons must measure: variation production, transmission, and ecological value as separate stages, with novelty defined relative to a declared comparison set.

2. Prior work and the experimental question

Sayama's original evoloop establishes interaction-mediated genotype change and spatial selection in deterministic, synchronous nine-state dynamics (Sayama, 1999). Later work shows that loops of the same size can differ in gene arrangement, colony morphology, persistence, and mutational bias. Salzberg and Sayama report round-robin competition among 15 size-4 genotypes and descendants inheriting variation biases (Salzberg and Sayama, 2004). These findings already connect inherited organization with evolutionary outcomes.

Salzberg, Antony, and Sayama examine genealogy, biased accessibility of larger species, pathogens, and localized or distributed disturbances (Salzberg, Antony, and Sayama, 2004). They also retain transient species that produce stable replicators in a later cycle. A failure to copy the first observed candidate immediately is therefore insufficient grounds for discarding its evolutionary potential. The recent review places these mechanisms within the broader problem of reproduction supporting inherited variation and discusses persistent questions of individuality and long-term evolution (Sayama and Nehaniv, 2025; preprint 2024).

Our experiment builds on these results by comparing the fate of variation under specified encounters. Population competition measures which types persist; here the immediate outcome is whether a particular encounter produces a new identity with a recoverable transmission path. Retaining nonproductive encounters supplies the denominator, while isolation tests and delayed extraction distinguish candidate appearance from later reproduction. The contribution is this intervention-and-measurement comparison, rather than the discovery of interaction-generated mutation or inherited mutational bias.

3. Methods

3.1 Dynamics and organisms

We use the original nine-state evoloop transition system, synchronous updates, and periodic boundaries. The neighborhood is center, top, right, bottom, left; rotations of explicit rules and the dissolution/default rules produce all 59,049 local transitions. The lookup table agrees exhaustively with the pinned CellPyLib reference and an independent expansion of the author's EvoLoopS rules (Antunes, CellPyLib; Sayama, EvoLoopS). Seed reproduction, deterministic replay, and checkpoint/resume are verified. Exact versions, hashes, licensing, and execution details are documented in source review (../docs/source_review.md) and reproduction instructions (../docs/reproducing-results.md).

S13 is CellPyLib's documented species13 seed, matching the author's 13.img, rather than the alternative default 13-2.img (Antunes, CellPyLib documentation). N01 is a stable descendant obtained from a 160×160 exploratory torus initiated with S13 at (65,45). Discovery samples every ten updates, considers up to four detached phases per candidate, and ends after two verified identities. This selected exploratory history supplies an experimental input; it is not replication across independently evolved populations.

A prospective discovery schedule (protocol 005) supplies the fresh extension with three naturally generated channel identities. It reverifies the original native N01 component, then searches three declared histories: the original 160-square S13 ecology to 12,000 updates, an S13/N01 encounter in 320-square space to 16,000, and an N01/N01 encounter in 320-square space to 16,000 or the three-identity stopping point. Sampling is every 20 updates, with at most two complete-component hashes per source composite identity and 32 verification trials per history. The first detached offspring must transmit its composite identity in two further isolated generations, and its word must equal the native source word. Selection also requires one unchanged core at elapsed offsets 20, 50, 80. Third detached descendants supply uniform input snapshots. Discovery selects on provenance and reproduction before viewing encounter outcomes; it is not independent-population replication.

3.2 Crossed encounters and controls

Before acquisition, protocol 001 specifies 32 main encounters, eight held-out encounters, and controls. Complete input snapshots advance alone by 0, 30, or 60 updates. Each resulting state must retain one recognized core, unchanged identity, and no boundary contact. These offsets refer to different snapshot origins; they are not necessarily homologous reproductive phases.

All four ordered S13/N01 pairings are tested against four main geometries at offsets 0 and 60, giving eight configurations per pair. Two held-out geometries at offset 30 give two further configurations per pair. Near configurations occupy 384×384 tori, with specified positions, rotations in 90-degree increments, and nonoverlapping input rectangles. Forty separated controls retain identities, offsets, orientations, and row displacement in 640×640 tori, with horizontal displacement of 320 cells. Twenty-four solitary controls cover organism, offset, and orientation. Every configuration runs for 1,600 completed updates; separated controls are checked for contact throughout.

The fresh protocol 006 defines nine ordered V01/V02/V03 pairings, four main configurations per pair at offsets 20/80, and two held-out configurations per pair at offset 50. Complete phase snapshots, relative displacements, and rotations differ from the original panel. All 144 near, separated, and solitary arenas use the same 640-square periodic domain. The 54 paired separated controls change only horizontal partner displacement; 36 solitary states cover each organism, offset, and orientation. The panel and its complete starting states were frozen before acquisition. Main and held-out configurations use different phases,

orientations, and offsets; each is a specified finite panel rather than a random sample of encounters.

3.3 What counts as a variant?

A structural detector samples rectangular zero-valued cores with intact inner sheaths in states 2/3 every ten updates. For channels containing only states 0/1/4/7, we read a clockwise signal word, preserving carrier spacing and normalizing by cyclic shift without reflection. Pairing this word with sorted core dimensions defines the operational composite identity. It preserves chirality but differs from the published counterclockwise, birth-anchored species identifier (Salzberg, Antony, and Sayama, 2004). Contaminated channels have unknown identity; recognized cores alone do not establish viability.

In the original atlas, a candidate composite identity must differ from both encounter inputs. A reporting audit introduced during acquisition, before inheritance analysis, additionally distinguishes signal-word novelty from shape-only differences. That composite endpoint remains the frozen primary definition; word-only novelty is a separately labeled, post-hoc sensitivity endpoint (protocol 002). The fresh atlas prospectively screens for a native signal word different from both inputs and records morphology-only changes separately. A candidate may match the third catalogue identity: novelty here is relative to the input pair, not global sequence novelty. This narrower screen leaves morphology-only routes to later word changes outside the inheritance endpoint.

Occupied input cells receive provenance masks 1 and 2; newly occupied cells inherit the union of occupied contributors in their neighborhood. Contact is the first update containing exclusive-provenance adjacency or mixed occupied provenance. These masks overapproximate causal contribution and do not identify genetic parents. After contact, novelty is consequently a pair-level outcome.

Core coordinates track structures. Appearances and losses are interval-censored within ten updates; reported candidate times are first sampled observations, not exact mutation or birth times. A changed word at an existing core can be registered without a new-core appearance event. The legacy detector does not join structures across the torus seam, so boundary contact limits the earlier native counts. The fresh atlas uses independently validated periodic recognition: four cuts cover cores whose six-cell padded bounding dimensions are smaller than half each domain dimension, occupied components are joined across seams, and a single-core component is unwrapped only when an empty row and column allow its complete extent to be retained. Winding and multicore components remain ineligible. Nonrectangular structures can escape either detector. Structure persistence, lineage survival, candidate novelty, inheritance, and benefit remain distinct measurements.

3.4 Testing transmission after isolation

The initial assay extracts the entire occupied connected component, including reproductive arms, if it contains exactly one recognized core and avoids the boundary. It places the component in an empty 192×192 torus. Shared, boundary-spanning, and oversized components are ineligible. The earliest two distinct extractable snapshots per run/identity define the search budget; size checks occur after capture, so oversized attempts consume that budget and remain unresolved.

Each generation runs for up to 1,600 updates. In the first legacy stage, the first newly positioned, valid closed core with a detached component supplies the next isolated generation. Later legacy stages select the first detached core matching the proposed

stable composite identity, skipping other identities; the expanded search instead retains alternate valid descendants. A candidate may first produce a different stable descendant. That descendant must transmit the same identity twice consecutively, requiring an additional generation when the original candidate was transient. Closure and detachment are recorded separately. Stable identities first arising after extraction are labeled assay-derived and checked again against both inputs.

A path without qualifying reproduction by the horizon is unresolved; an empty grid is extinction under isolation. Either outcome can coexist with an unexamined successful path or ecological dependence. Complete-component extraction preserves active arms, avoiding the disruption caused by cropping only the central body.

3.5 Delayed extraction and expanded descendant search

Protocol 004 tests whether maturation or a broader search recovers inheritance missed by the initial assay. It follows all seven candidate identities, nested within six originating configurations. At first observation, the lexicographically first matching core tuple fixes the target. No later successful structure may replace a lost target. Verified S13 and N01 snapshots in empty 384×384 worlds provide positive controls.

Extraction occurs after 0, 60, 120, or 240 updates: 28 candidate-delay and eight control observations. Ten-update tracking distinguishes target loss and subsequent same-tuple recurrence. Lost, recurrent, unknown, shared, boundary, and oversized targets are ineligible. Each delayed world also continues natively for 1,600 updates. Matching words elsewhere are structural observations; they do not establish offspring ancestry. These continuation windows overlap and are not independent observations.

Every eligible component enters both the unchanged legacy assay and an expanded search. The expanded search collects the first three detached, newly positioned valid cores per generation, once per core tuple, ordered by time, coordinates, and snapshot hash. This is the operational offspring criterion; a core appearance event alone is not sufficient. Depth-first traversal permits at most six simulated generations along a path and 24 generation simulations per root, each lasting at most 1,600 updates. Identity changes reset the transmission streak. Two identical consecutive transmissions confirm stability. Candidate recovery requires a stable signal word different from both inputs; control recovery requires the original composite identity.

Comparing searches on the same snapshot isolates search sensitivity. Comparing delays combines maturation with continued ecological interaction. Ineligible extractions and unexamined branches remain explicit.

The fresh atlas applies the same expanded search budgets with periodic recognition to the earliest two distinct complete-component snapshots per run/composite candidate identity. Stable transmission requires two consecutive identical composite identities and a stable word different from both inputs. Transients may produce another stable identity. Size-ineligible snapshots consume an attempt; missing components and capped paths remain unresolved. Native appearance, isolated transmission, and ecological dependence are separate outcomes.

3.6 Analysis and ecological benefit

The original primary endpoint is any confirmed novel stable composite identity per near configuration; the fresh primary endpoint is any confirmed stable word different from both inputs, conditional on its prospective native-word screen. Each retains all declared near states in its denominator: 40 in the original atlas and 54 in the fresh atlas. Rates are

reported separately because the panels and candidate screens differ. Catalogue-relative novelty is a supplementary description, not a replacement for the pair-relative endpoint. Candidate and unresolved outcomes are reported separately. For each main or held-out panel, an interaction contrast is a difference of differences; the original two-input comparison is:

[rate(S13,S13) − rate(N01,S13)] − [rate(S13,N01) − rate(N01,N01)].

This measures pair-configuration interaction under ambiguous parentage, rather than an identified focal mutation capacity. The configurations form a specified finite panel, not a random sample from a natural encounter distribution. Deterministic reruns verify reproducibility. Accordingly, we report descriptive contrasts without binomial intervals or inferential p-values. Held-out geometries assess configuration transfer; they do not supply independent organisms or evolutionary histories. For the three-input atlas we report focal and partner marginal means, residuals from their additive decomposition, and all nine contrasts obtained by choosing two focal identities and two partner identities. A strict rank reversal requires opposing, nonzero focal contrasts across partners; persistence requires the same two signs in the held-out panel. These analyses concern pair-level outcomes because provenance does not resolve genetic parentage.

The ecological schedules and benefit criterion were frozen with protocol 006 before the confirmed outcomes were known. Protocol 007 prospectively specifies state preparation, the fixed score registry, score calibration, and arithmetic for the paired comparison. Both acquisition and analysis are complete; the following describes the implemented score and paired criterion. A separate audit examines whether reserved conditions supply a new ecological challenge.

3.7 Ecological score and recovery calibration

We retain both inherited outcomes. RV01 is the recovered V01 composite identity from the V02/V03 encounter; RV02 is the 14×14 identity absent from the three-input catalogue, from V01/V03. Their representatives are chosen by earliest native capture followed by run/root order, using the complete final detached descendant. V01/V02/V03 retain their third detached discovery snapshots. No arms are stripped and no preparation phase is advanced. RV01 and V01 have the same composite identity but different complete snapshots, so these are comparisons of declared states rather than words alone.

Each whole snapshot is rotated, then placed in an empty 192×192 torus with its top-left row and column at 96 − floor(height/2) and 96 − floor(width/2). All five states and their hashes are retained. Their four unique exact composite identities define one fixed score registry, deduplicating V01/RV01. The score counts every recognized core with an identity in that registry, regardless of its starting organism. Unregistered valid and unknown-identity cores are recorded separately; the registry is never expanded in response to performance. This is a known-identity structural population measure, potentially omitting viable unregistered descendants.

The prospective recovery calibration comprises five baseline snapshot controls and ten new ecologies: each prepared state under rotation 0/block (80,80) and rotation 1/block (80,112). These conditions differ from the development and reserved confirmation conditions. At native updates 400, 620, 1,020, and 1,400, recognized cores are divided into registry-scored and unscored classes; the latter includes unregistered valid and unknown identities. The first and last distinct lexicographic core tuples are selected within each class, once if only one exists, with an explicit empty stratum if none exists. Selection precedes inspection of descendant fate. At most 80 scored and 80 unscored observations are selected; they are nested deterministic observations, not independent

organisms or histories.

Eligible complete single-core components retain arms and use validated periodic unwrapping. Shared, winding, oversized, and otherwise ineligible components remain in the selected-observation denominator. The bounded isolation search retains up to three detached valid offspring per generation, six generation states in depth, and 24 generation simulations per root, each with a 1,600-update horizon. Scored observations must recover two consecutive transmissions of the originally counted exact composite identity, allowing transient intermediates. Unscored observations can recover any stable composite identity and provide a descriptive measure of excluded recoverable structures. Baselines require recovery of their original identity.

The calibration gate requires all five baselines, at least 90% original-identity recovery among all selected scored observations, at least 80% in each starting-state group with observations, and at least one recovery in every nonempty state-by-condition-by-source-time scored stratum. A starting-state group without selected scored observations, or a zero overall denominator, fails the implemented gate. These fixed finite-panel thresholds are practical validation criteria; ineligibility and nonconfirmation do not prove native nonviability.

Each selected native source also receives a 1,600-update structural continuation, reusing its ecology extended to update 3,000. Remaining absolute disturbances at 600/1,000 are retained, with none thereafter. Fixed-core loss/recurrence, endpoint identity, and matching words elsewhere are recorded at 20-update resolution. The windows overlap and do not assign offspring ancestry. Even a passing gate supports sampled recoverability under imposed isolation and a calibrated known-identity proxy, not an exact census of viable organisms in the crowded native world.

3.8 Frozen paired ecological comparison

All five prepared states undergo all four frozen conditions: development rotations 0/1 with block top-left coordinates (40,40)/(128,128), and reserved confirmation rotations 2/3 with coordinates (40,128)/(128,40). Each condition runs for 1,600 updates, clearing a 24×24 block at 600 and 1,000. The unchanged synchronous CA advances to the disturbance time, the block is set to state zero, then detection and sampling occur. The 81 samples at 0,20,…,1,600 include the starting state and all zero counts with equal weight. Actual removed occupied cells and pre/post core counts are retained, including zero-impact removals; these cannot by themselves establish recolonization after population loss.

Five states across four conditions yield 20 prepared state-condition simulations. Common input scores are reused across pairs. RV01 is compared conservatively with V02 and V03; RV02 with V01 and V03. Within each condition, candidate and both inputs receive the same schedule. Each panel averages its two condition scores. The strict criterion requires a candidate to exceed each originating input in both development and confirmation panels; exact ties fail. Comparison uses integer count sums over 162 samples per panel, avoiding rounding artifacts. Conditions are deterministic assessment configurations, not independent population replicates.

If calibration passes, the criterion records a finite advantage under the calibrated known-identity population proxy. Its intended ecological interpretation concerns persistence and recolonization; actual depletion and distinct confirmation challenges must be checked separately. If it fails, every one of the 20 frozen states still runs as a structural diagnostic, while benefit confirmation is blocked and beneficial-heritable yield remains unknown. Revised scores or gates require a separate prospective protocol. Encounter denominators remain 36 main, 18 held-out, and 54 overall, with other candidate

fates unresolved. Neither a structural score increase nor confirmed assay advantage alone establishes increased evolvability, H1/H3, or recursive self-improvement.

4. Results

4.1 A transient structure supplies an inherited input variant

In discovery, a candidate first observed at update 2,910 had an 11×13 core and a signal word different from S13. Isolation yielded a 12×12-core descendant, followed by two generations transmitting that stable identity. Detachment was first observed after 380, 390, and 390 updates in the successive assays, at ten-update resolution. N01 denotes the stable descendant. The transient and stable forms share the changed word, distinguishing morphological stabilization from signal change. This observed route motivates allowing transient-to-stable paths in encounter assays.

The prospective discovery reached three eligible identities in 79 selected native-component trials: the original N01 source and 78 new trials. V01 is the reverified N01 word; the native sources of V02 and V03 were first observed in the N01/N01 history at updates 3,440 and 5,040, sampled every 20 updates. Each retains its native word while stabilizing from a 12×14 to a 13×13 core under isolation. Their stable words differ from one another and from V01. Seventy-four trials were unresolved or changed identity before completing the declared transmission chain; two recovered stable words different from their native source and were therefore rejected. These counts describe input selection, not encounter yields.

Input	Native history, time	Native → stable core	Detachment times
V01	Original discovery,2910	11×13 →12×12	380,390,390
V02	N01/N01,3440	12×14 →13×13	430,410,420
V03	N01/N01,5040	12×14 →13×13	410,420,410

Times are updates from each isolated generation's start, sampled every 10. Both new identities come from the same history and are separately verified, not independently evolved population replicates. These positive transmission paths broaden the input panel and show that the earlier negative encounter result does not amount to a general inability of transients to stabilize.

4.2 Encounters produce candidate changes without confirmed stable transmission

All 40 near configurations contacted within the observation window. Five of 32 main configurations and one of eight held-out configurations produced candidate composite changes. All seven candidate identities also differed in signal word from both inputs, so the post-hoc word-only classification agrees here. Separated controls produced 0/40 candidates and solitary controls 0/24. No atlas state received a boundary-touch flag. The matched near-minus-separated candidate contrast is 6/40, or 0.150, for this panel.

Five candidate-producing configurations supplied 11 isolation snapshots; the remaining candidate could not be separated. None confirmed stable novel transmission. Six paths produced changed detached descendants in their first stage: four subsequently dissolved, and two failed to reproduce the same identity by the next horizon. Two initial snapshots dissolved, and three produced no qualifying detached descendant. These observations separate candidate appearance from transmission under the stated assay.

Pair	Candidates: main	Candidates: held-out	Inherited: main; held-out
S13 × S13	3/8	0/2	0/8; 0/2
S13 × N01	0/8	0/2	0/8; 0/2
N01 × S13	1/8	1/2	0/8; 0/2
N01 × N01	1/8	0/2	0/8; 0/2

Changing the focal organism in the main panel gives candidate contrasts of +0.250 with S13 as partner and −0.125 with N01 as partner: a rank reversal and interaction contrast of +0.375. Held-out contrasts are −0.500 and 0.000, giving −0.500 without a strict rank reversal. The candidate pattern does not transfer to the held-out configurations. All confirmed-inheritance contrasts are zero because no tested path satisfies the criterion; the candidate pattern therefore does not establish an interaction in useful heritable variation.

Thirty near states lost both initial recognized cores, although many retained input-type descendants. Five ended without a recognized core: one empty and four retaining debris. Initial cores also disappeared in all separated and solitary controls while colonies continued. Individual-core loss is therefore an unsuitable substitute for lineage extinction.

4.3 Neither delayed extraction nor broader search recovers candidate inheritance

An exploratory extension first followed the earliest N01×N01 offset candidate, observed at encounter update 1,550. Its 3×12-core source produced a 6×6 descendant, then a 4×4 descendant that dissolved at the third isolation stage. This selected diagnostic motivates examining more paths, but adds no independent confirmation.

The subsequent prospective calibration completed all 36 observations. Of 28 candidate-delay observations, 15 were eligible, ten had lost the fixed source core, and three remained shared. No extraction was excluded for recurrence, unknown identity, boundary contact, or size. All eight controls were eligible. Both searches confirmed 0/15 novel candidate paths and recovered 8/8 known control identities. None of the six originating configurations supplied a confirmed novel path. The 13 ineligible observations remain in the denominator for extraction eligibility; their inheritance outcomes are missing.

Delay	Eligible / observed	Novel: legacy; expanded	Parents: legacy; expanded
0	4/7	0/4; 0/4	2/2; 2/2
60	4/7	0/4; 0/4	2/2; 2/2
120	4/7	0/4; 0/4	2/2; 2/2
240	3/7	0/3; 0/3	2/2; 2/2

Expanded search examined 69 candidate and 16 control generation states. No root reached the 24-simulation or six-transmission limit. Eighteen candidate generation states reached the three-offspring collection limit, leaving later offspring unexamined. Each delay of the 3×12 source generated ten searched states without stable recovery. Other eligible roots produced descendants that dissolved under further isolation or remained unresolved at the horizon.

In native continuation, the 9×9 candidate word appeared at another core at original encounter times 1,150 and 1,160. Its appearance in all four overlapping windows is one structural recurrence, not four independent reproduction events. No candidate original word was recognized at the endpoint of its continuation windows. However, boundary flags occurred in 16/28 candidate and 2/8 control windows, limiting these counts. Control words

appeared at new locations and persisted at all eight endpoints. Native endpoint absence consequently cannot establish candidate extinction.

4.4 A fresh panel recovers two pair-relative inherited outcomes

EXP-atlas-002 completed all 144 frozen states. Contact occurred in all 36 main and 18 held-out near configurations, at updates 151–990 and 251–1,090 respectively. None of the 54 separated controls contacted. Native candidate words occurred in 14/36 main and 4/18 held-out states, compared with 0/54 separated and 0/36 solitary states. No morphology-only changed identities were observed in near states. The paired near-minus-separated candidate contrast is 18/54, or 0.333.

Pair	Candidates: main	Candidates: held-out	Inherited: main	Inherited: held-out
V01 × V01	1/4	0/2	0/4	0/2
V01 × V02	2/4	0/2	0/4	0/2
V01 × V03	2/4	0/2	1/4	0/2
V02 × V01	2/4	1/2	0/4	0/2
V02 × V02	0/4	0/2	0/4	0/2
V02 × V03	2/4	1/2	1/4	0/2
V03 × V01	1/4	1/2	0/4	0/2
V03 × V02	2/4	0/2	0/4	0/2
V03 × V03	2/4	1/2	0/4	0/2

Bounded serial isolation confirms pair-relative inherited novelty in 2/36 main configurations and 0/18 held-out configurations. Combined yield is 2/54, or 0.037, conditional on this panel, horizon, screen, and search. Fourteen main and four held-out candidate-producing states supply 52 run/composite identities and 88 extraction roots. Eight roots confirm the endpoint, but they converge on two stable identities from two configurations; they are not eight independent inherited events. Six candidate identities lack an extractable phase. The other 16 candidate-producing configurations remain unresolved for stable novel inheritance under the bounded search.

Both successful configurations use the parallel geometry at elapsed offset 20. In near-main-V01-V03-p20-a-parallel, contact begins at update 224. A 13×15-core candidate observed at 1,110 produces a 14×14-core isolated descendant, followed by two transmissions first observed after 440 updates each. Its stable word equals the native candidate word and differs from all three catalogue words. A matching 14×14 composite identity appears natively at 1,550 and also reproduces twice after extraction. Roots captured ten updates apart converge on identical later snapshots. This is one configuration with a stable word absent from the catalogue, giving supplementary catalogue-relative yield 1/36 main and 0/18 held out.

In near-main-V02-V03-p20-a-parallel, contact begins at update 233. A 12×13 candidate at 460 produces a different word in a 12×12 isolated descendant, which transmits twice, first observed after 390 updates in each generation. The stable identity exactly matches V01. That word is absent from V02 and V03 and therefore satisfies the frozen pair-relative endpoint; it is a recovery of an existing catalogue identity. V01's word also appears natively at 860 in a 12×12 core and reproduces after extraction. Its native appearance does not assign ancestry to the earlier candidate. The changed-word transition along the 460 extraction path occurs in the imposed isolation assay; the stable-word native observations and their own isolation tests provide separate evidence.

Figure: Two recorded routes to stable transmission (see appended figure pages).

Each row shows one complete captured component and its recorded isolated descendants from

a successful configuration. Multiple extraction roots within each encounter converge on the same stable identity; they are not independent encounter outcomes. Sources and regeneration commands are in the figure manifest.

The expanded search executes 1,205 generation simulations. It records 134 simulation-budget and 24 depth-limit truncations; 441 nodes reach the three-offspring collection cap. Later branches, unsampled phases, and ecological dependence remain unexamined. Edge activity occurs in 73/144 native arenas; the fresh periodic detector handles validated small rectangular bodies and contractible single-core components, while large, nonrectangular, winding, and shared structures remain outside the confirmed extraction scope.

Main-panel candidate contrasts show two strict focal rank reversals. For V01 minus V02, the contrasts are −0.250 against V01 and +0.500 against V02, producing a difference of differences of −0.750. For the same-size V02 minus V03 comparison, the contrasts are +0.250 and −0.500, producing +0.750. Neither strict reversal persists held out. Inherited main contrasts compare V01 or V02 against V03: changing from a V01 or V02 partner to V03 changes the focal contrast from zero to +0.250, giving a difference of differences of −0.250. All held-out inherited contrasts are zero, and neither split contains an inherited strict rank reversal. These are descriptive pair-configuration interactions; the concentration of successes in the V03 partner column alone would not establish H2. Both inherited outcomes enter the subsequent paired ecological comparison.

4.5 Sampled recovery passes the frozen ecological-score calibration

EXP-ecological-calibration-001 completed five baseline controls and ten calibration ecologies. All five baselines recovered their original composite identity. All 68 selected registry-scored components were eligible; 63 recovered their original identity through the required stable transmission chain, and five remained unresolved under the bounded search. The pooled recovery fraction is 63/68 (92.6%). Each starting-state group exceeds its 80% threshold, and all 40 nonempty scored strata contain at least one recovery. The frozen calibration gate therefore passes.

Prepared state	Original-identity recovery / selected
V01	14/15
V02	11/12
V03	12/13
RV01	14/15
RV02	12/13

The calibration executes 141 generation simulations; 136 nodes reach the three-offspring collection cap, with no depth or simulation-budget truncations. No unscored core is selected because all sampled unscored strata are empty. Recovery sensitivity for viable unregistered identities therefore remains untested; empty strata do not establish the completeness of the fixed registry.

Each of the five nonconfirming roots supplies no detached qualifying child within its isolated horizon. Three isolated grids become empty (V02, V03, and RV02), while V01 and RV01 remain nonempty without qualifying reproduction. These are three extinctions under the imposed isolation assay and two unresolved reproduction outcomes, not five demonstrated native deaths. In every corresponding native continuation, the fixed-coordinate core is lost and later recurs with the original identity, present again at the continuation endpoint. Recurrence at the same coordinates does not establish that the same individual survived or assign offspring ancestry.

Figure: Recovery calibration for the fixed population registry (see appended figure pages).

The original identity is recovered from 63/68 selected scored components and 5/5 baselines. Components are nested within ten deterministic ecologies; they are not independent population histories. Unscored-class recovery is untested because no such observation is selected. Stratum-level results (../figures/eco1-calibration-strata-dark.svg) retain the finite denominators.

4.6 One inherited state meets the finite proxy criterion, without independent ecological confirmation

EXP-benefit-001 completes all 20 frozen state-condition simulations. The score counts cores in the fixed four-identity registry. V02 has an unregistered valid core at two of 81 sample times in each condition; unknown-identity cores also occur briefly in other states. These structures are excluded from the score and their viability remains untested. Empty unscored strata at the calibration source times do not imply absence of unscored structures in the paired worlds. Each panel combines 162 equally weighted samples from two conditions. Development and reserved confirmation means are identical for every prepared state:

Candidate	Originating inputs	Candidate mean	Input means	Frozen criterion
RV01, recovered V01	V02, V03	3.580	3.370, 3.321	Met
RV02, new 14×14	V01, V03	3.136	3.562, 3.321	Not met

RV01 exceeds V02 by 34/162 counted-core sample units (+0.210) and V03 by 42/162 (+0.259) in each panel. RV02 falls below V01 by 69/162 (−0.426) and V03 by 30/162 (−0.185). No tie or rounded comparison determines these outcomes. The formal proxy-criterion yield is 1/36 main configurations, 0/18 held out, and 1/54 overall. It belongs to the known-word recovery, not the catalogue-novel word. The other 16 candidate-producing encounters remain unresolved under the earlier inheritance search. This finite classification does not turn their missing outcomes into failures. The evidence-stage figure (../figures/eco1-evidence-stages-dark.svg) separates confirmed transmission, the operational proxy criterion, and unresolved independent ecological benefit.

Figure: Scores in all four frozen ecological conditions (see appended figure pages).

Counts concern registered closed cores, not an exact viable-population census. All four schedules ran for every state; reserved schedules supply the same score trajectories as development under rotation. Paired margins (../figures/eco1-paired-margins-dark.svg) compare each candidate conservatively against both inputs.

Actual exposure strongly limits the ecological claim. Of 40 scheduled block clearings, 36 remove no occupied cells. Four remove 18 cells each at update 1,000, affecting only V01 and RV01 in development condition 1 and confirmation condition 0. All four leave the registered-core count unchanged at four. Thus no clearing demonstrates immediate depletion of the measured population. No paired undisturbed controls were included, preventing separation of baseline multiplication from a response to disturbance.

A symmetry audit further shows that reserved confirmation condition 0 reproduces development condition 1 under a 90-degree rotation, and confirmation condition 1 reproduces development condition 0 under a 270-degree rotation, with periodic centering translations where required. Initial and final complete states and all 81 score samples match these transformations. The only effective block removals rotate with the states; the others are no-ops on occupied cells. The reserved panel therefore provides no independent

ecological challenge. The recorded frozen criterion is met for RV01, while independent ecological benefit confirmation remains unresolved. These results do not demonstrate recolonization after population depletion or general ecological superiority.

Figure: Recorded states surrounding scheduled disturbances (see appended figure pages).

Literal cell-state snapshots show the imposed blocks and measured removals. Zero-impact clearings are retained as observations. Recorded animations and regeneration commands are indexed in the figure manifest; neither structural persistence nor a block schedule alone demonstrates recolonization.

5. Discussion

5.1 Production and transmission require separate explanations

The encounter comparison identifies configurations that produce changed signal words; the inheritance comparison asks whether those changes persist through repeated reproduction. The original atlas and calibration show divergence between these measurements; the fresh atlas recovers stable transmission in two configurations. Treating candidate counts as evolvability would nevertheless give the main-panel reversals a generality that neither atlas supports on held-out configurations. Confirmed transmission also distinguishes a word absent from the catalogue from recovery of a known third identity. The known-word recovery meets the frozen ecological proxy criterion, whereas the catalogue-novel identity does not; independent ecological benefit confirmation remains unresolved.

The calibration addresses two specific explanations for the negative inheritance result. Allowing the prescribed maturation delays and following additional detached descendants did not recover a stable novel path. Successful recovery of S13 and N01 shows that both searches can extract and transmit those tested identities at those delays. It makes a blanket explanation that the assay never recovers reproducing loops incompatible with the controls. It does not establish sensitivity for every altered phenotype.
Candidate-specific extraction damage, ecological assistance, later offspring, and longer transient sequences remain possible.

The discovery path supplies a useful contrast: a transient structure did produce stable novelty under complete-component isolation. It demonstrates a route accommodated by the method, while its exploratory selection prevents using it to estimate encounter-wide success. The fresh positive paths further demonstrate that transient candidates can lead to stably transmitting organisms. The current evidence supports conditional variation in production and recovered inheritance, followed by a finite population-score advantage for one prepared state. Recoverability calibration supports that proxy in its sampled conditions; it does not remove the exposure and symmetry limitations of the paired comparison.

5.2 Scope of the causal comparison

The original two-organism panel contains one selected descendant from one history. Signal word, shape, arms, and reproductive phase can covary; a pair effect cannot be assigned solely to a gene sequence. Equal observation horizons also permit unequal time after contact. Separated controls change world extent as well as distance, limiting interpretation to the declared configurations. The fresh controls hold arena extent fixed. V02 and V03 have the same 13×13 core dimensions, giving a comparison at common core size, but arms, signal words, complete states, and developmental timing still vary. All three fresh inputs are related, selected descendants; two derive from the same discovery history.

Isolation deliberately changes the environment and can disrupt or stabilize reproduction. Fixed-core tracking excludes lost or reshaped targets and their possible descendants. The three-offspring cap, ineligible phases, finite horizons, rectangular detector, and seam-crossing failures bound the searched paths. Negative outcomes identify what was not recovered under specified interventions, while positive outcomes identify the particular paths those interventions recover. Successful roots stop at their first accepted identity, so the search does not exhaust the number or diversity of viable variants within a productive encounter.

The ecological comparison adds limitations of measurement and exposure. Calibration establishes sampled original-identity recovery after isolation, not native reproductive fate for every counted body. The fixed registry can miss viable unregistered descendants; no unscored selections test that possibility here. RV01 and V01 share a composite identity but differ in their complete prepared snapshots, and their means differ by 3/162 counted-core sample units. Performance therefore cannot be assigned to the signal word alone. Different arms or reproductive phases, core size, and baseline multiplication may contribute. Most clearings miss all occupied cells, none immediately reduce the counted population, and the reserved conditions are rotational duplicates of development. The proxy result is reproducible, but it does not independently confirm adaptive value under a new disturbance challenge.

5.3 Connecting encounters, organisms, and history

The fresh atlas broadens the verified input panel and recovers positive paths under its declared endpoint. Its higher observed yield cannot be attributed to panel enrichment alone: inputs, domain size, native screening, periodic measurement, and descendant search differ from the original atlas. It is a new conditional comparison, not a controlled estimate of the effect of changing any one of those factors. No stable-yield contrast transfers to the fresh held-out states.

The next informative experiment uses prospectively calibrated, nonsymmetric impact-matched disturbances and paired undisturbed controls. Retain both inherited outcomes and both originating inputs, compare multiple declared reproductive phases or matched complete-state preparations, and reserve challenges that cannot be obtained by rotating or translating development states and schedules. Before outcome acquisition, verify that the disturbance policy removes its intended occupied structures or counted cores, and record actual absolute and fractional depletion. A state-dependent targeting policy, if used, must be declared as an intervention rather than treated as a shared fixed location. Repeat score calibration in the new depleted ecologies. Paired undisturbed runs separate baseline multiplication from disturbance response; the comparison remains conditional on its frozen metric and panel. The detailed plan is in the next-experiment plan (../docs/next-experiment.md).

H1 extends this comparison by sampling organisms from independently evolved populations and testing them, alongside matched ancestors, against a common held-out encounter panel. Higher beneficial-heritable-variant yield than matched ancestors under these shared conditions would locate an increase in transferable organism state. Population history supplies replication.

H3 crosses organisms and encounter distributions derived from undisturbed, structured-disturbance, and appropriately shuffled histories. Transplanting history-derived organisms into common encounters isolates the organism pathway; exposing shared organisms to history-specific encounters isolates the exposure pathway. Their cross identifies dependence between these contributions. In deterministic dynamics, history acts

through present state and scheduled interventions: identical complete states and schedules have identical futures.

Together these experiments turn the title's question into comparisons with interpretable outcomes. Their common requirement is to follow variation from its production through transmission to ecological consequence. The present study executes encounter, transmission, and paired population-score comparisons. It recovers two pair-relative inherited outcomes and a formal proxy advantage for one state, while exposing why independent ecological confirmation requires effective and distinct challenges. Production, transmission, and benefit cannot be collapsed into a single novelty count. H1 and H3 remain untested; no evidence of recursive self-improvement is claimed.

Reproducibility and references

Protocols, configurations, complete starting grids, compact evidence, analyses, and figure-generation scripts are available at https://github.com/ReloadLightly/evoloop-levels. Reproduction instructions (../docs/reproducing-results.md) link commands and evidence locations. Figures and their data provenance are indexed in the figure manifest (../figures/manifest.json). Exact reference versions, licensing, and bibliographic caveats are documented in the source review (../docs/source_review.md); verified bibliographic entries are in references.bib (references.bib).

Sayama, H. (1999). A New Structurally Dissolvable Self-Reproducing Loop Evolving in a Simple Cellular Automata Space. *Artificial Life*, 5(4), 343–365. DOI: 10.1162/106454699568818.

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Sayama, H., and Nehaniv, C. L. (2025). Self-Reproduction and Evolution in Cellular Automata: 25 Years After Evoloops. *Artificial Life*, 31(1), 81–95. DOI: 10.1162/artl_a_00451. Earlier version: arXiv:2402.03961v2 (2024).

Software sources: Luis M. Antunes, *CellPyLib* and its evoloop documentation (inspected 2026); Hiroki Sayama, *EvoLoopS*, version 1.0 (1999). Exact versions and source links are recorded in the bibliography and source review.

Verified references

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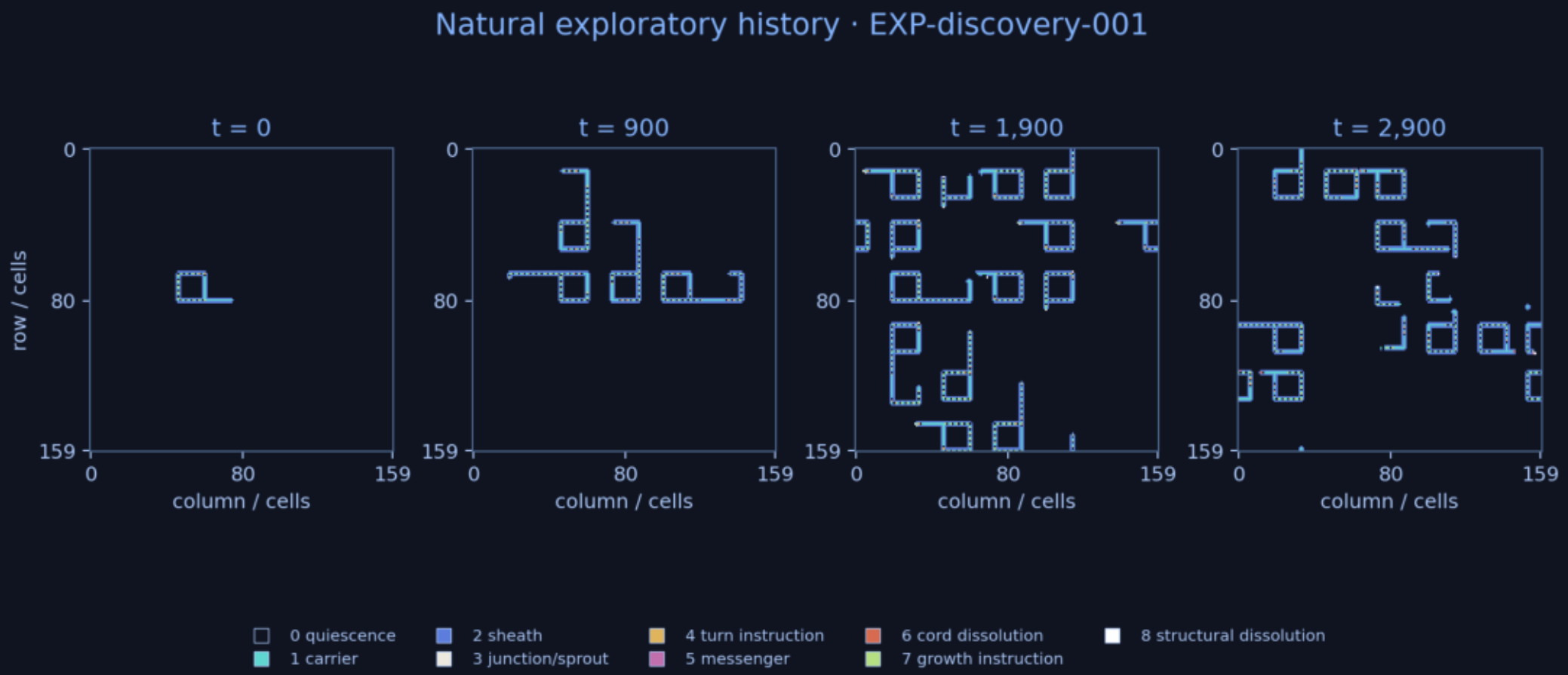


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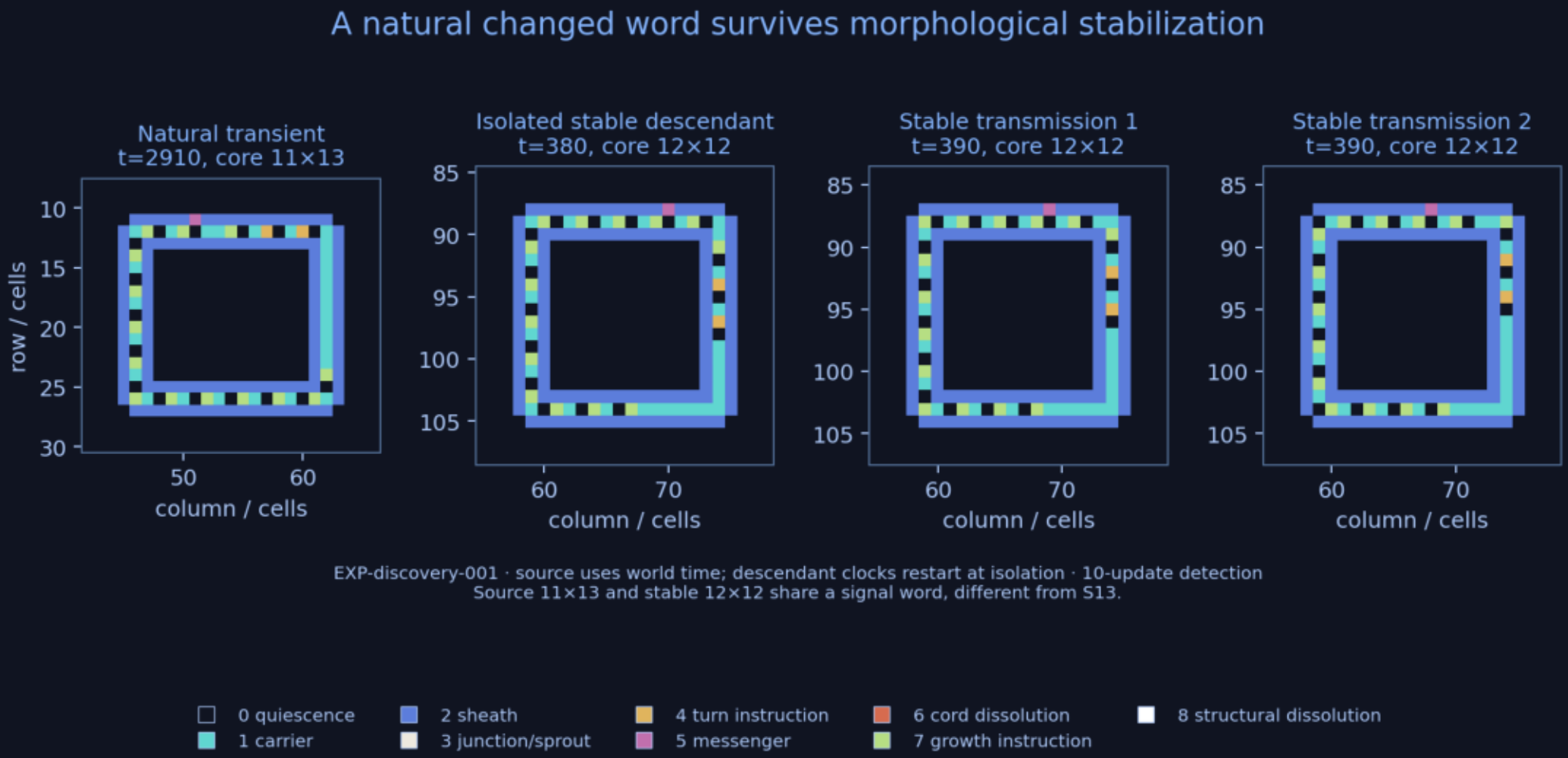


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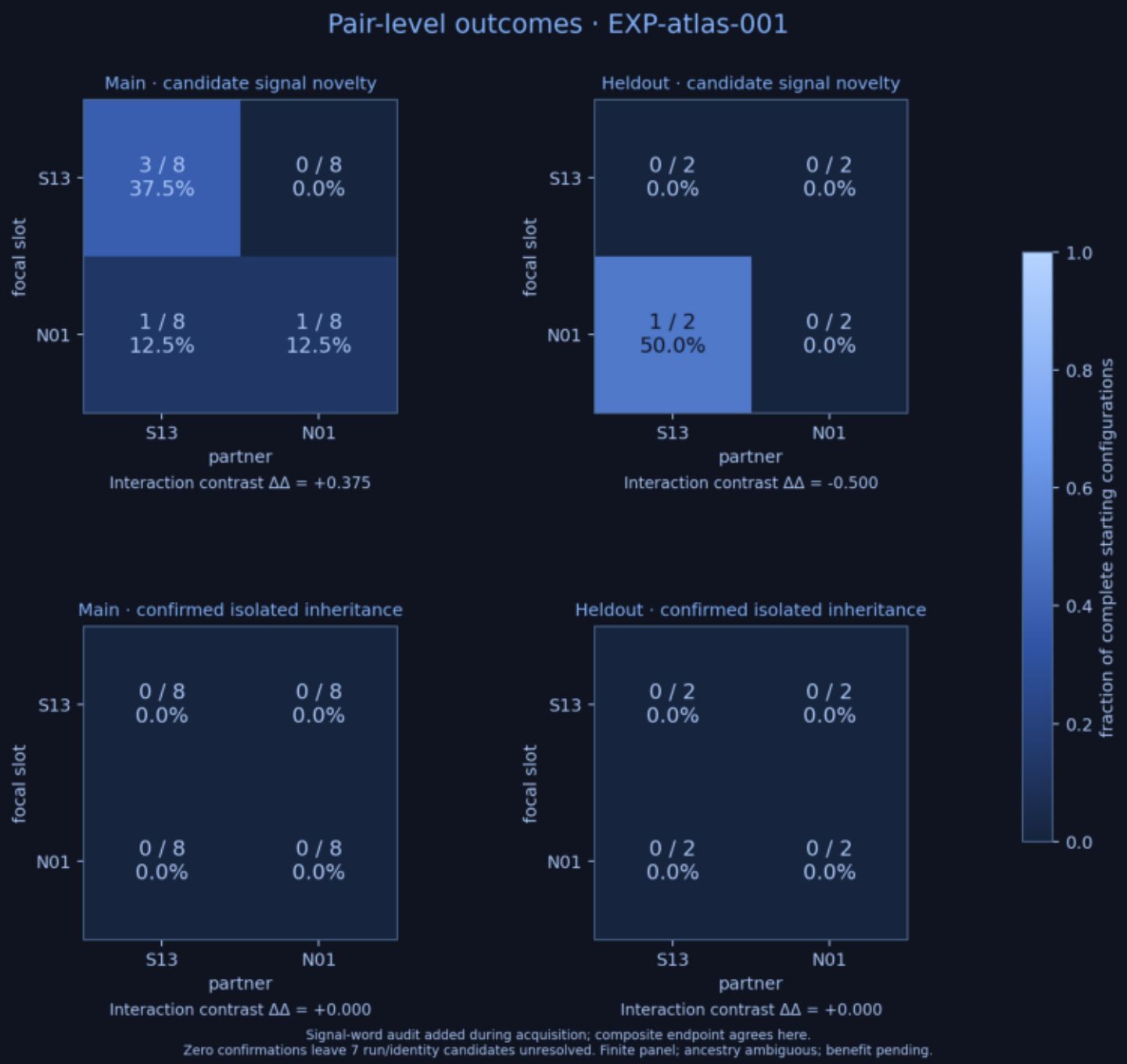


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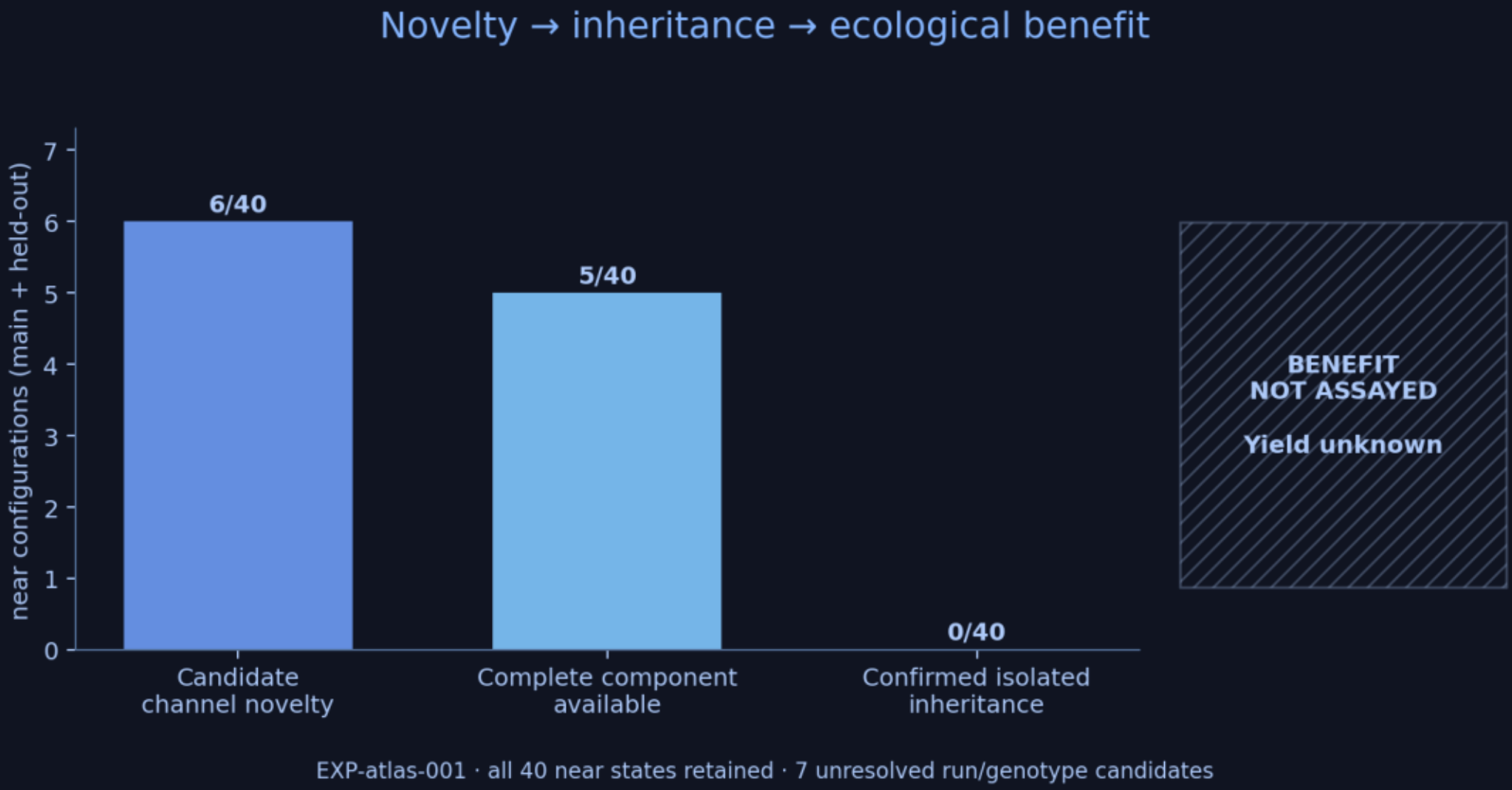


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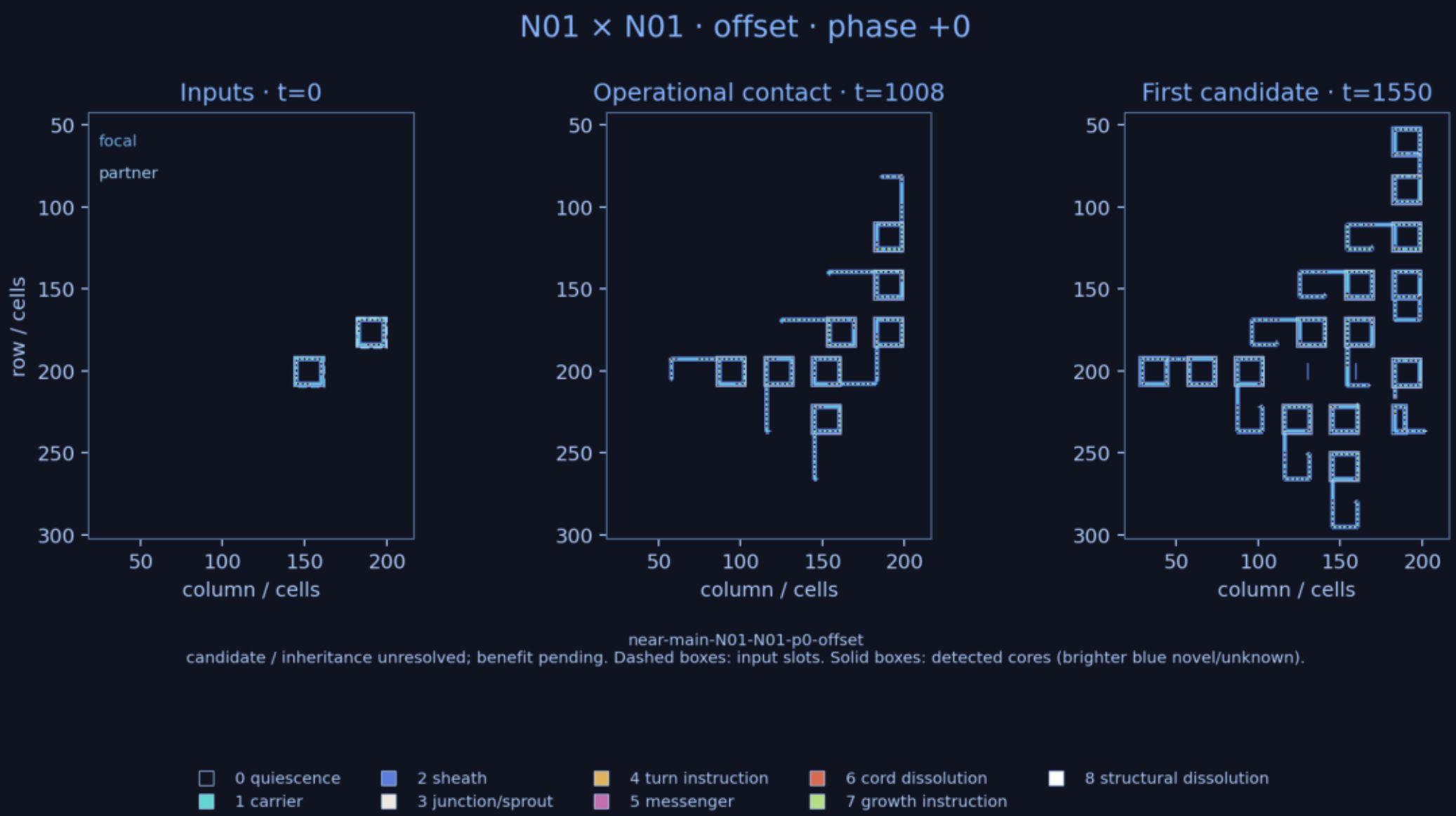


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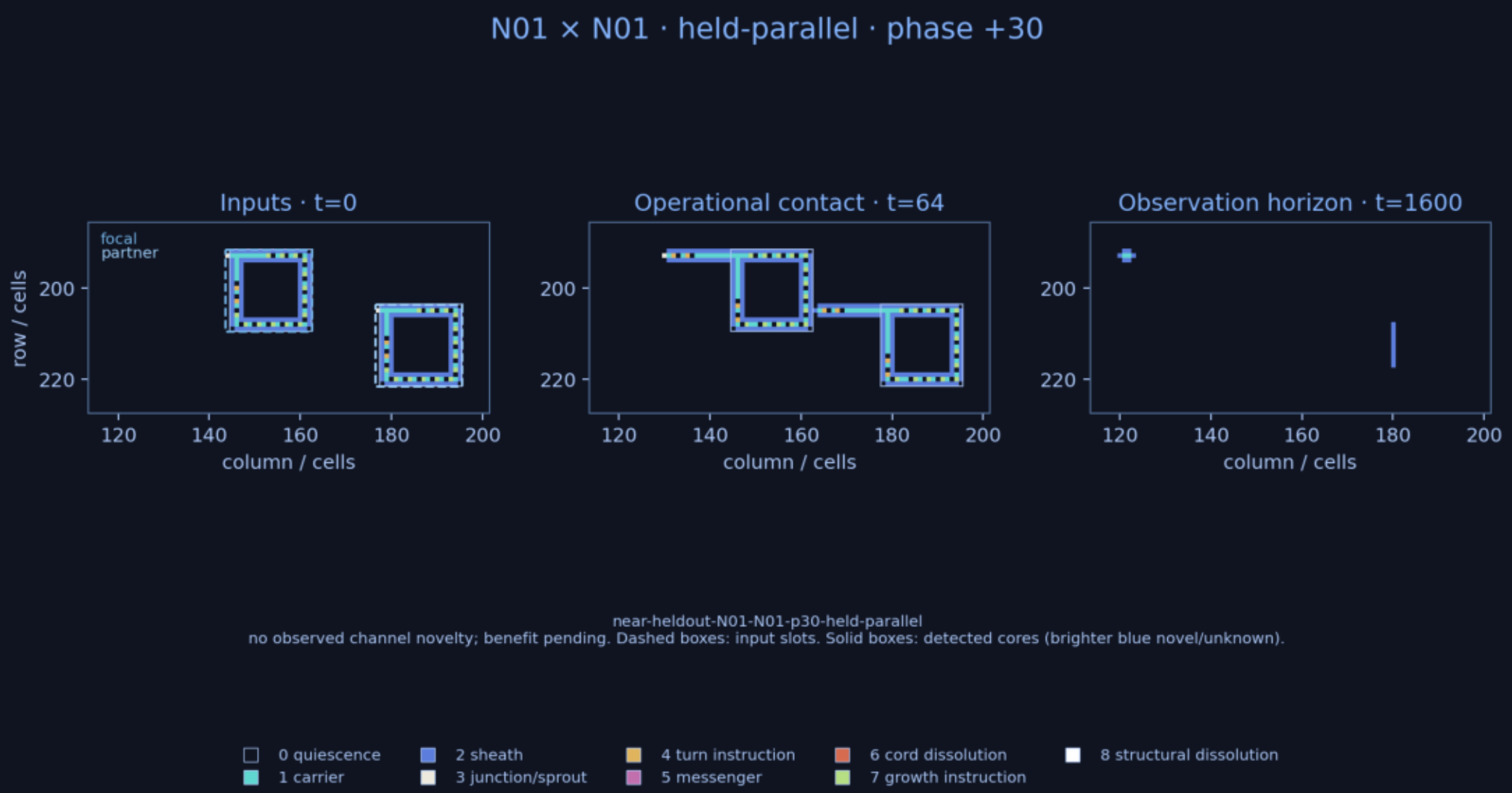


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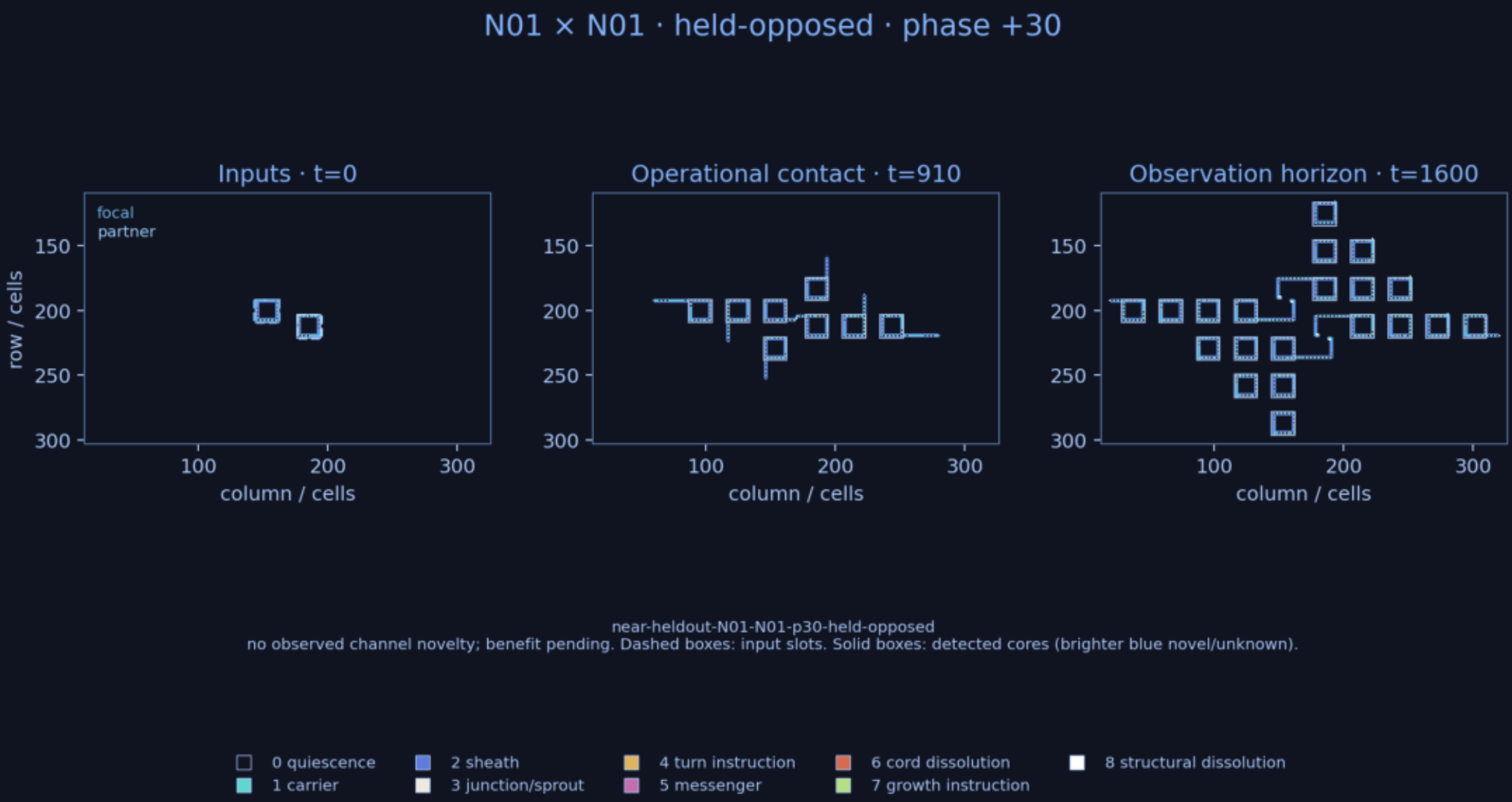


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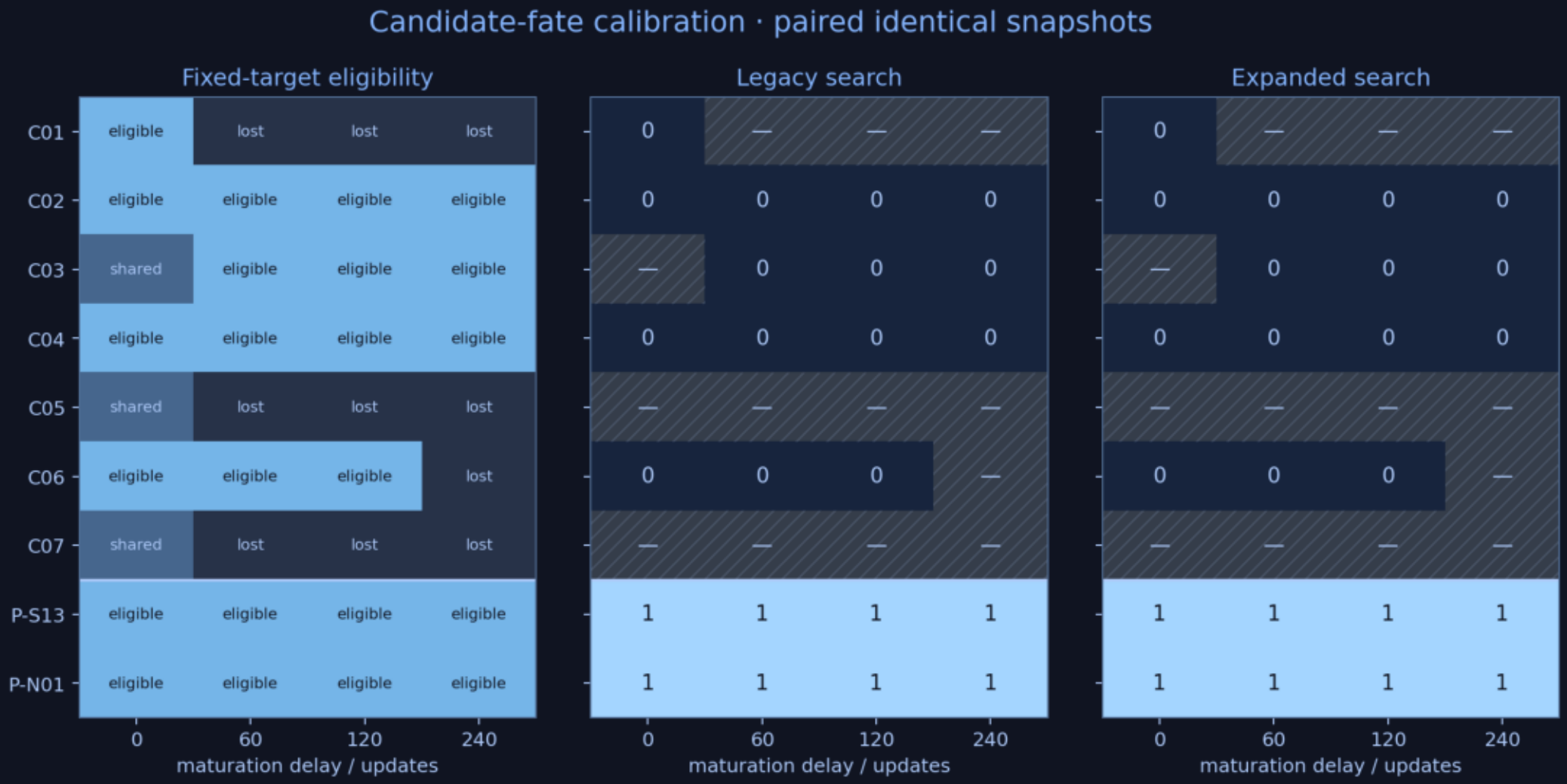


Figure: fate-calibration-matrix. Source and regeneration: figures/manifest.json.

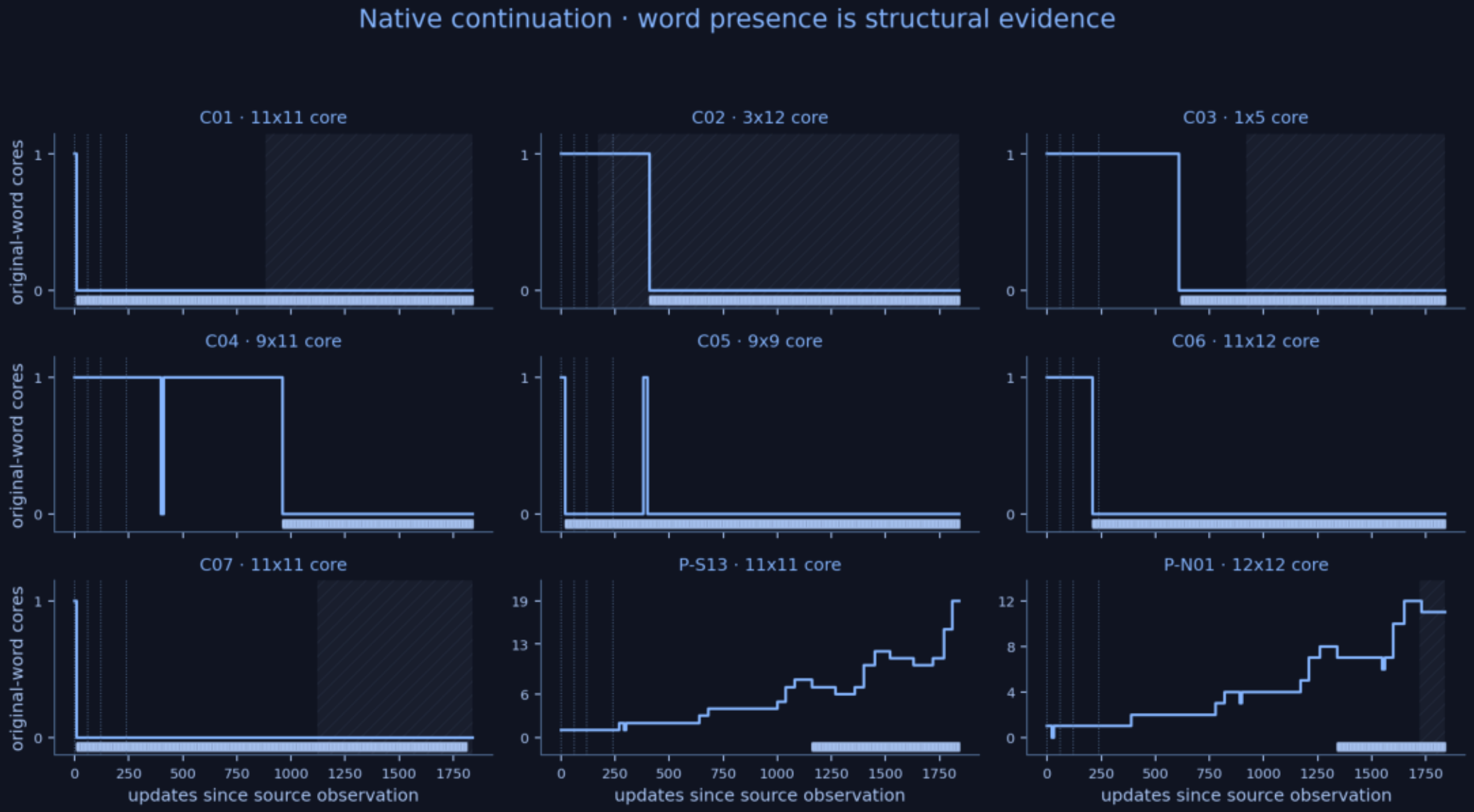


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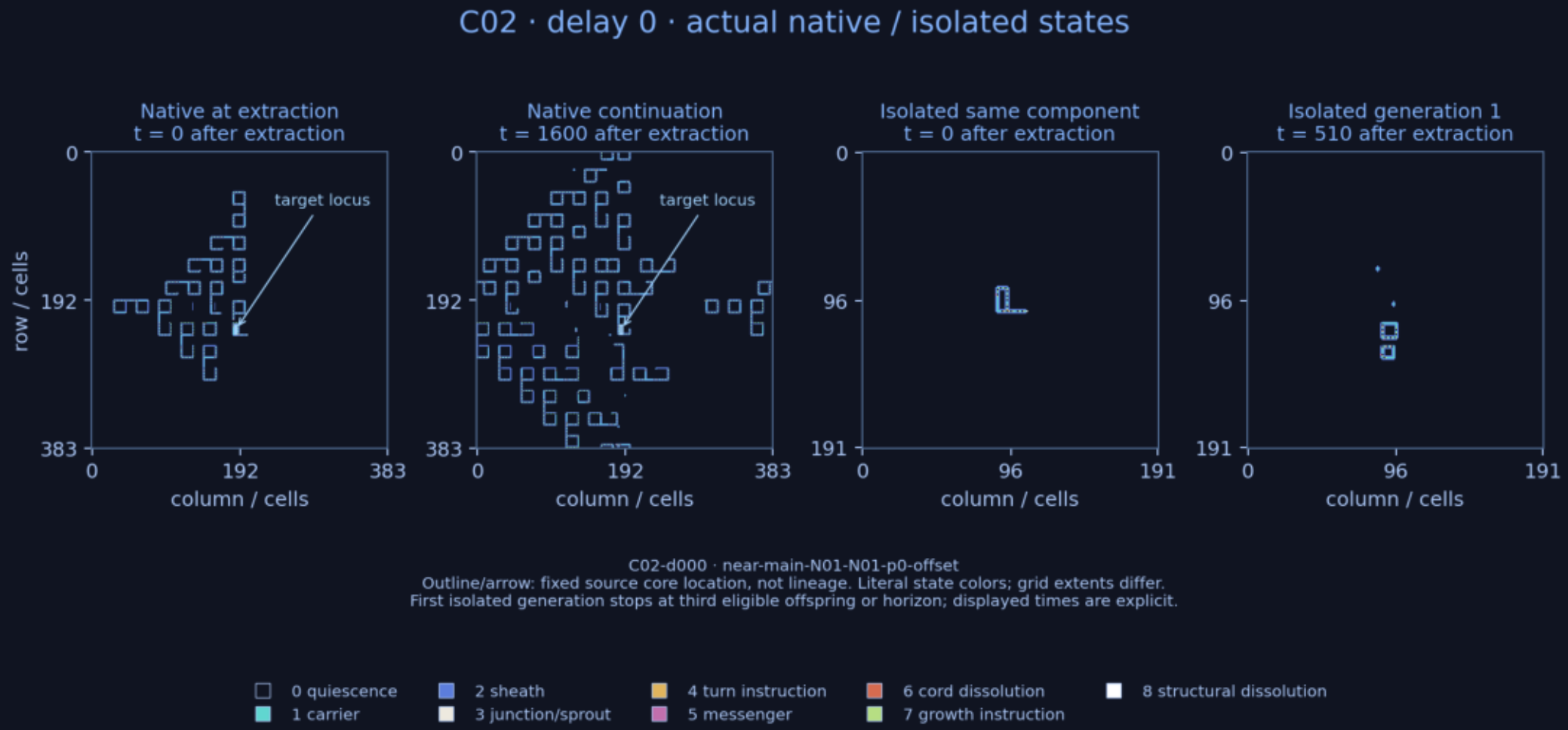


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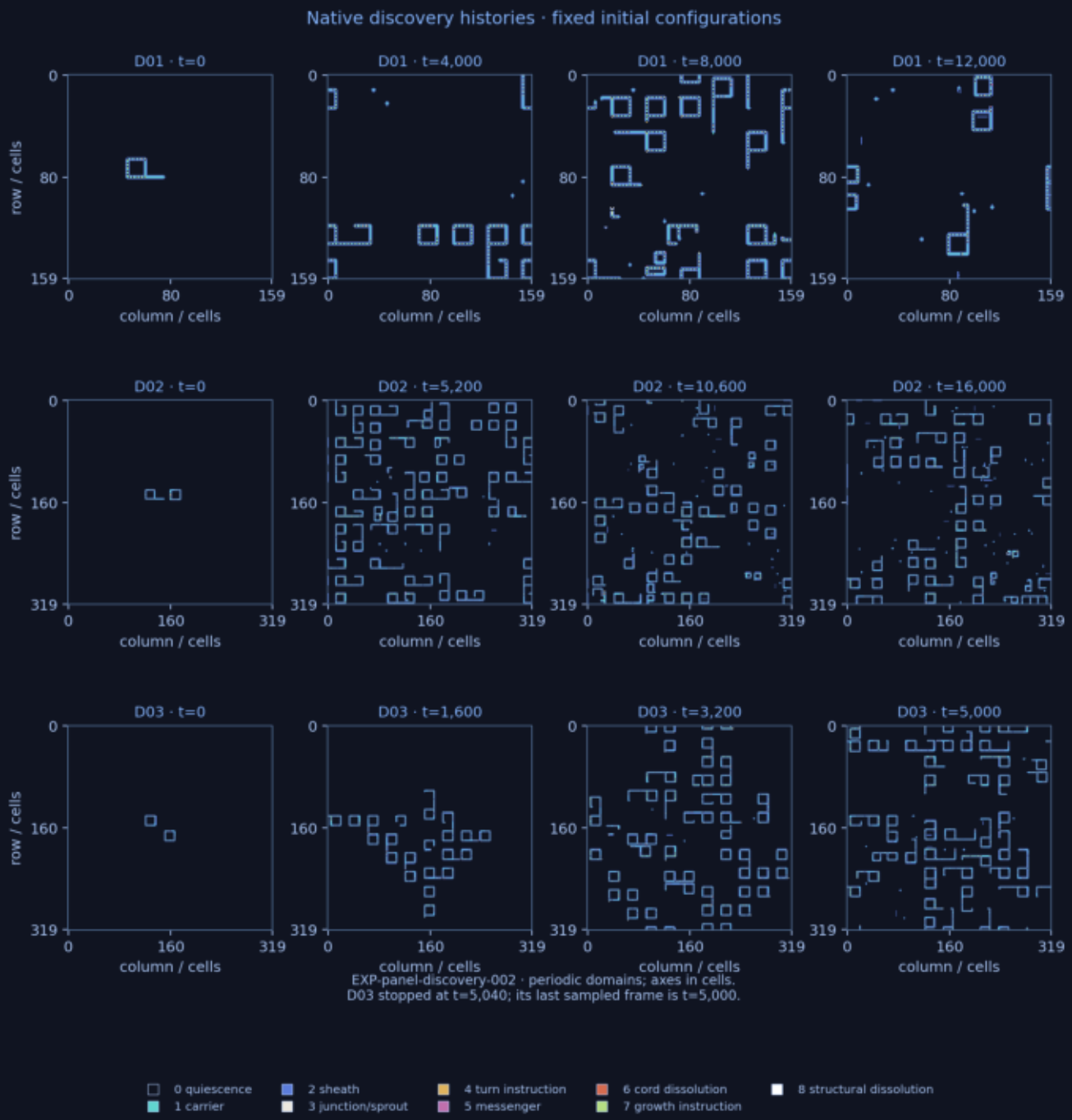


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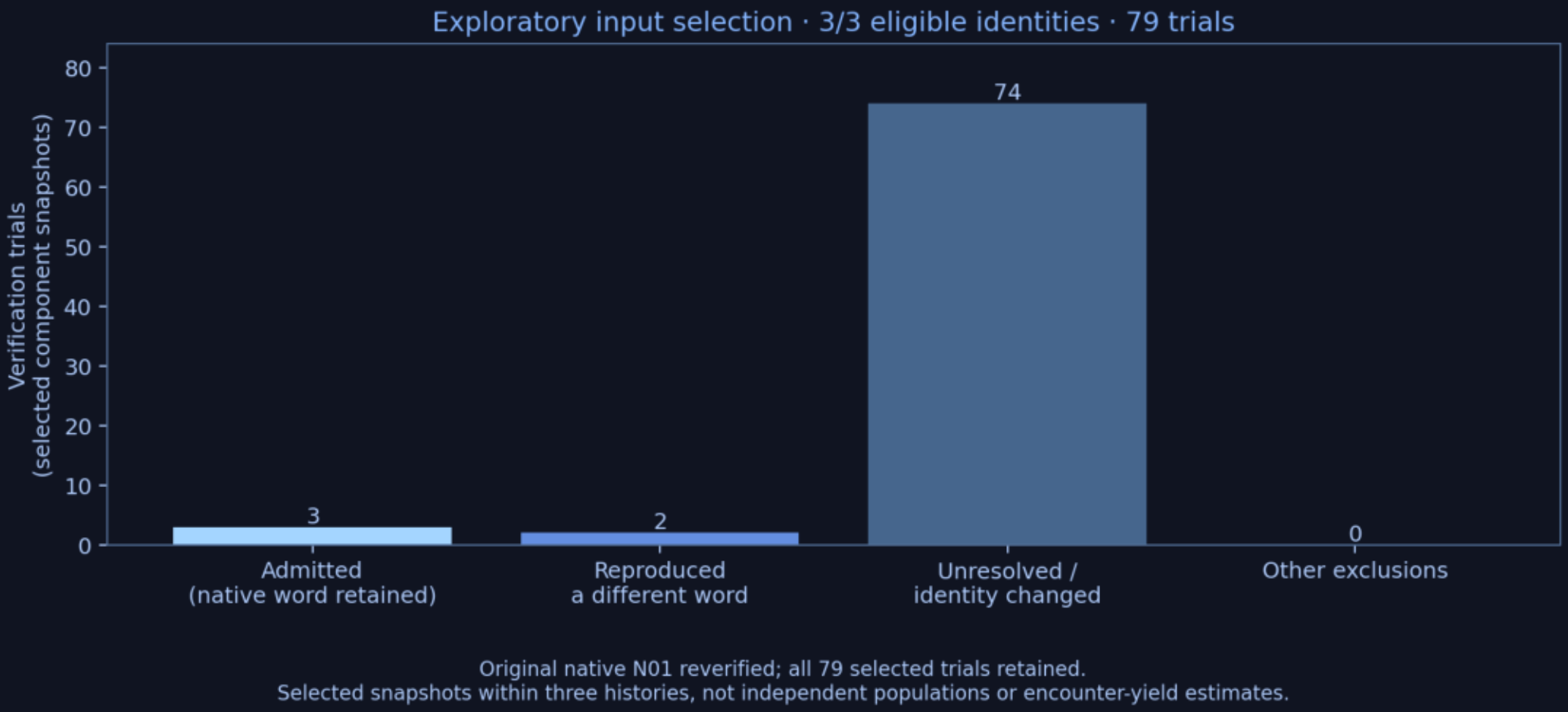


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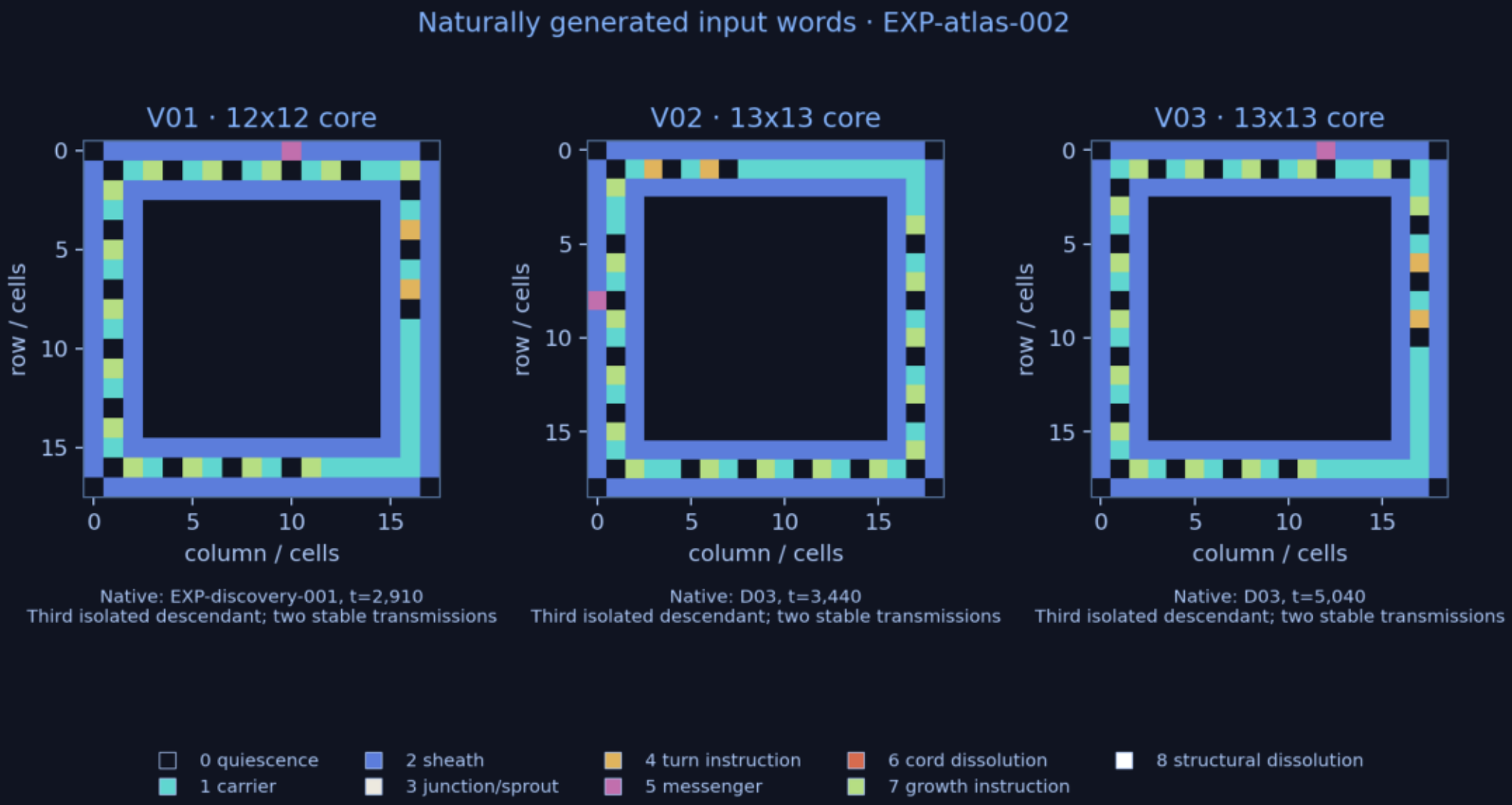


Figure: atlas2-verified-panel. Source and regeneration: figures/manifest.json.

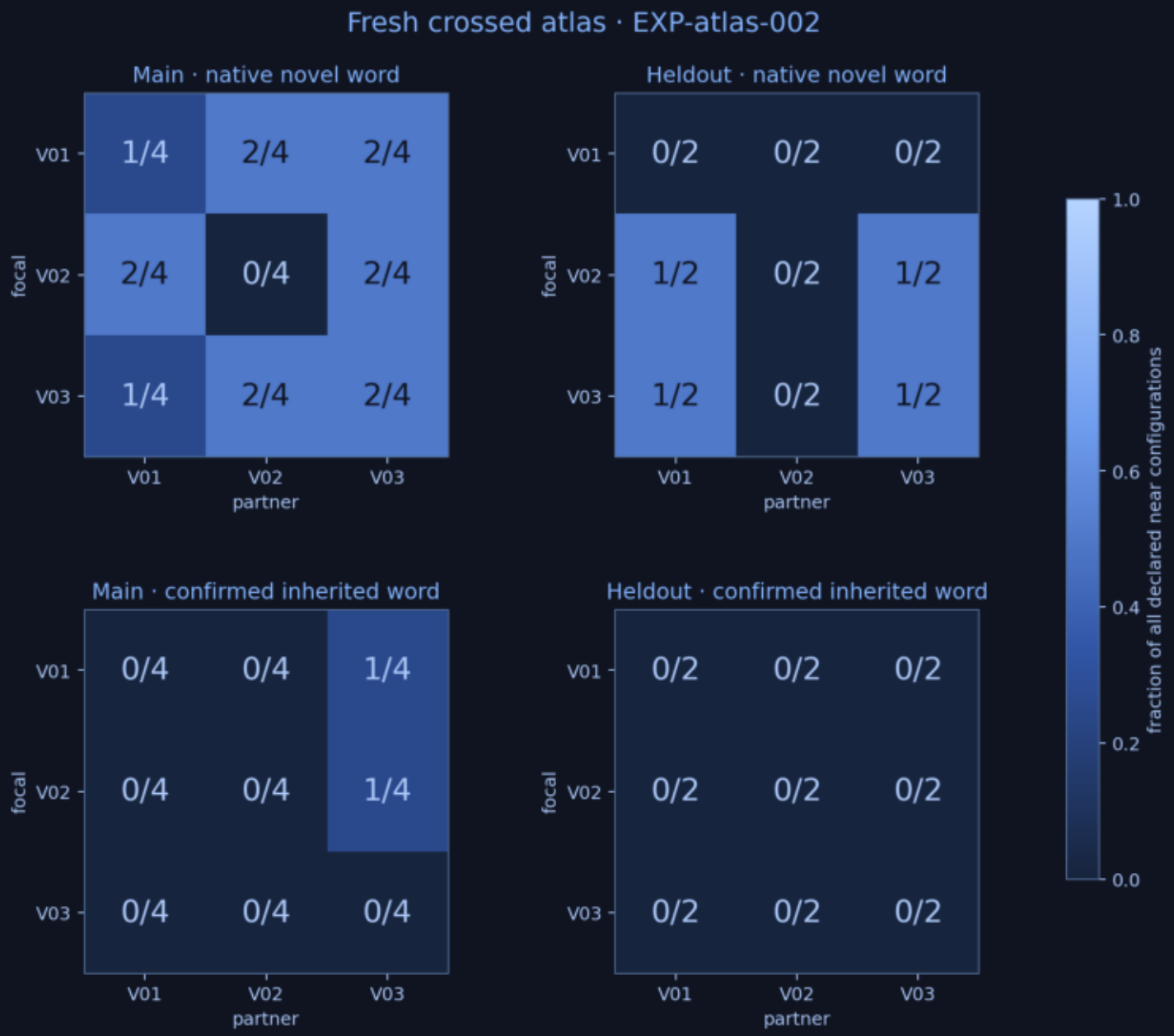


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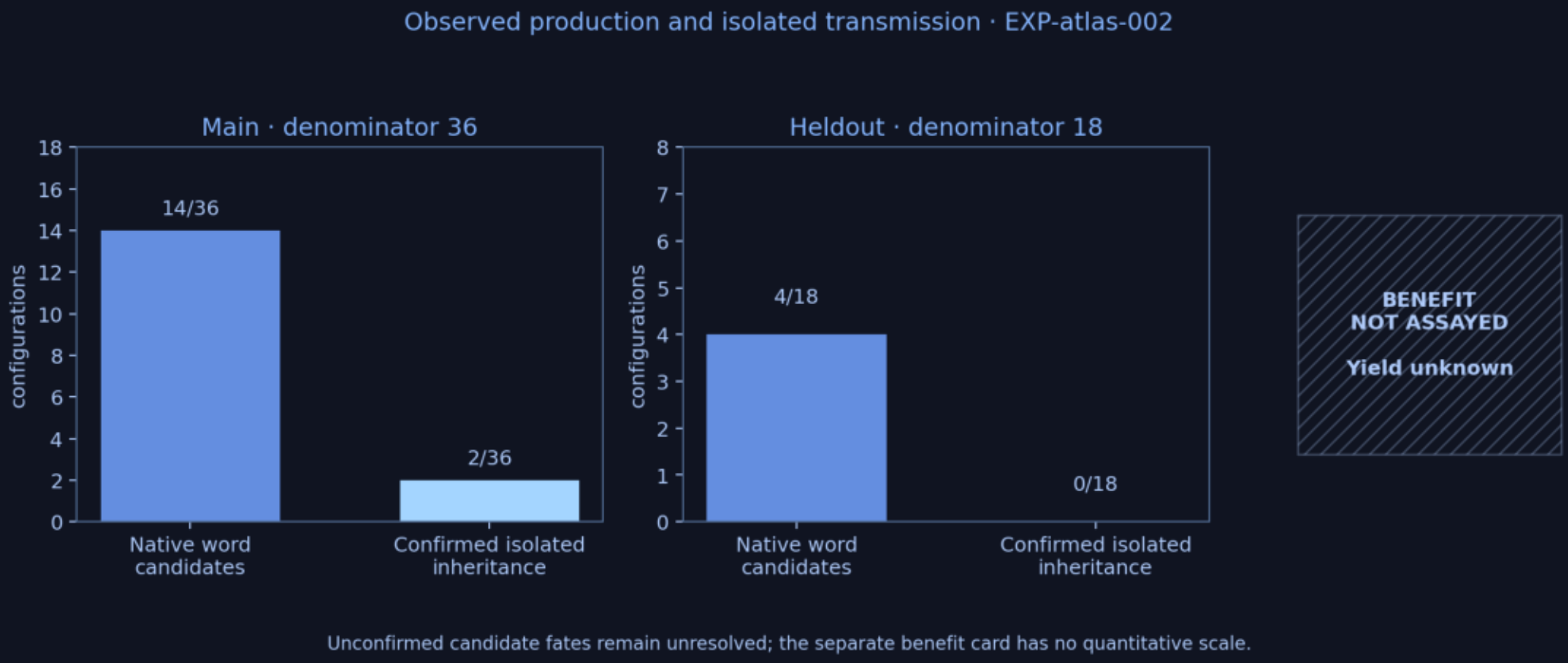


Figure: atlas2-evidence-stages. Source and regeneration: figures/manifest.json.



Figure: atlas2-inheritance-path. Source and regeneration: figures/manifest.json.

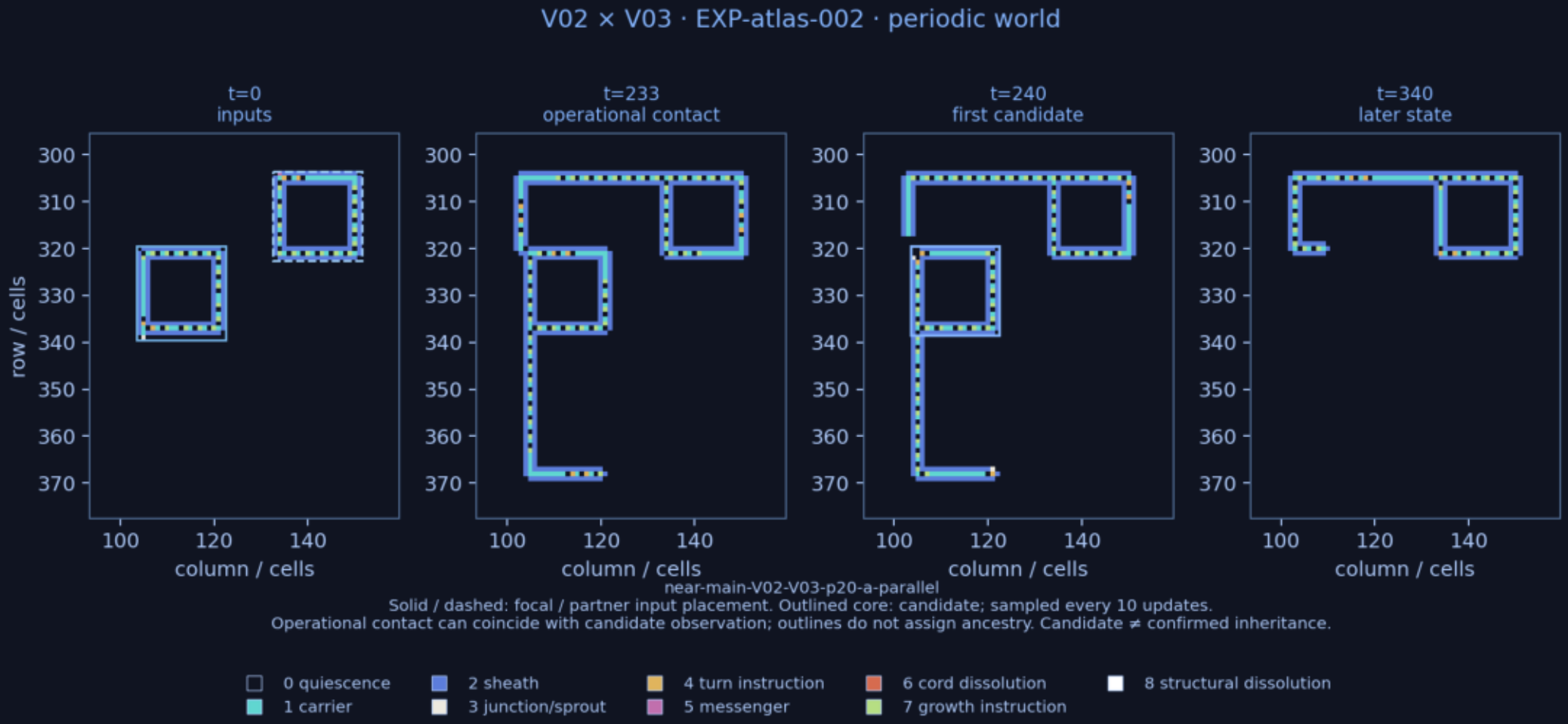


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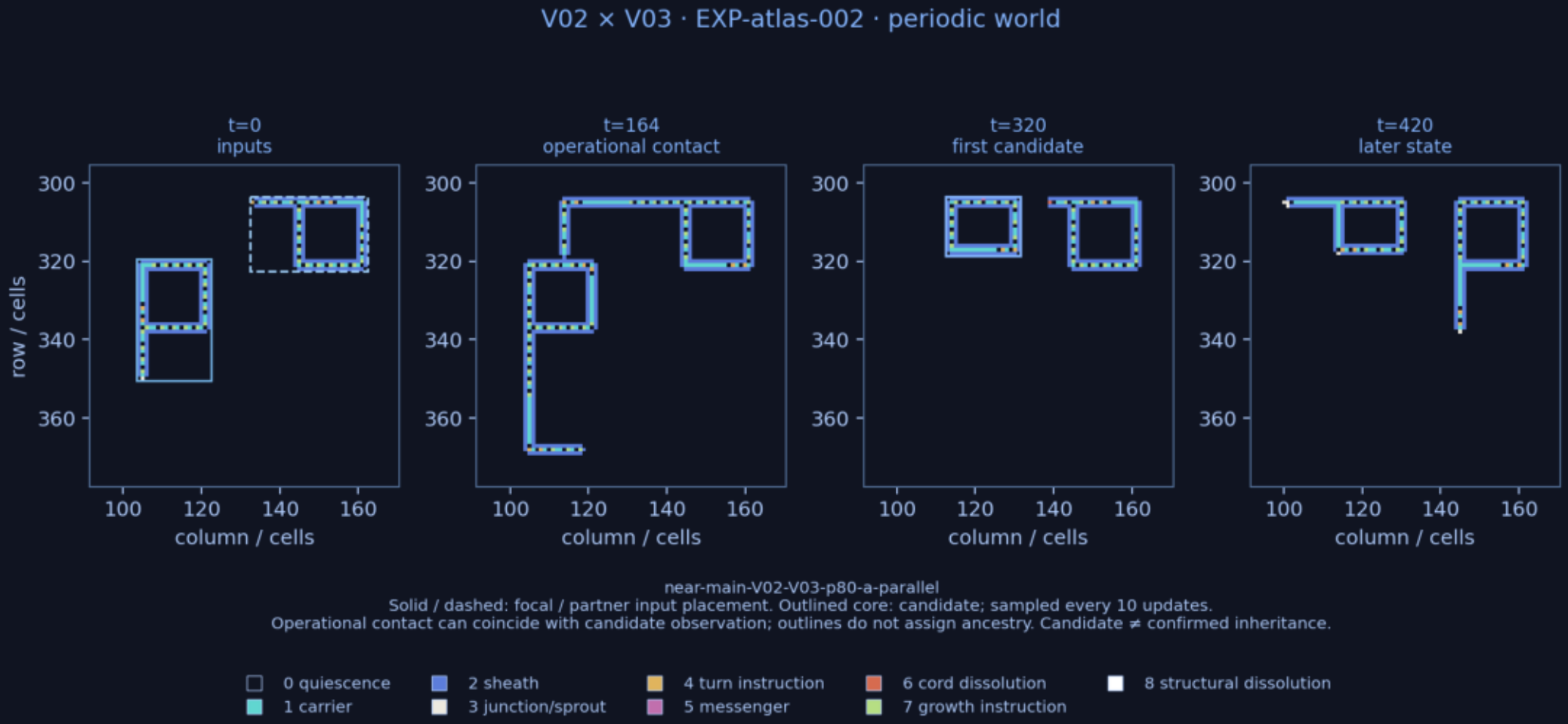


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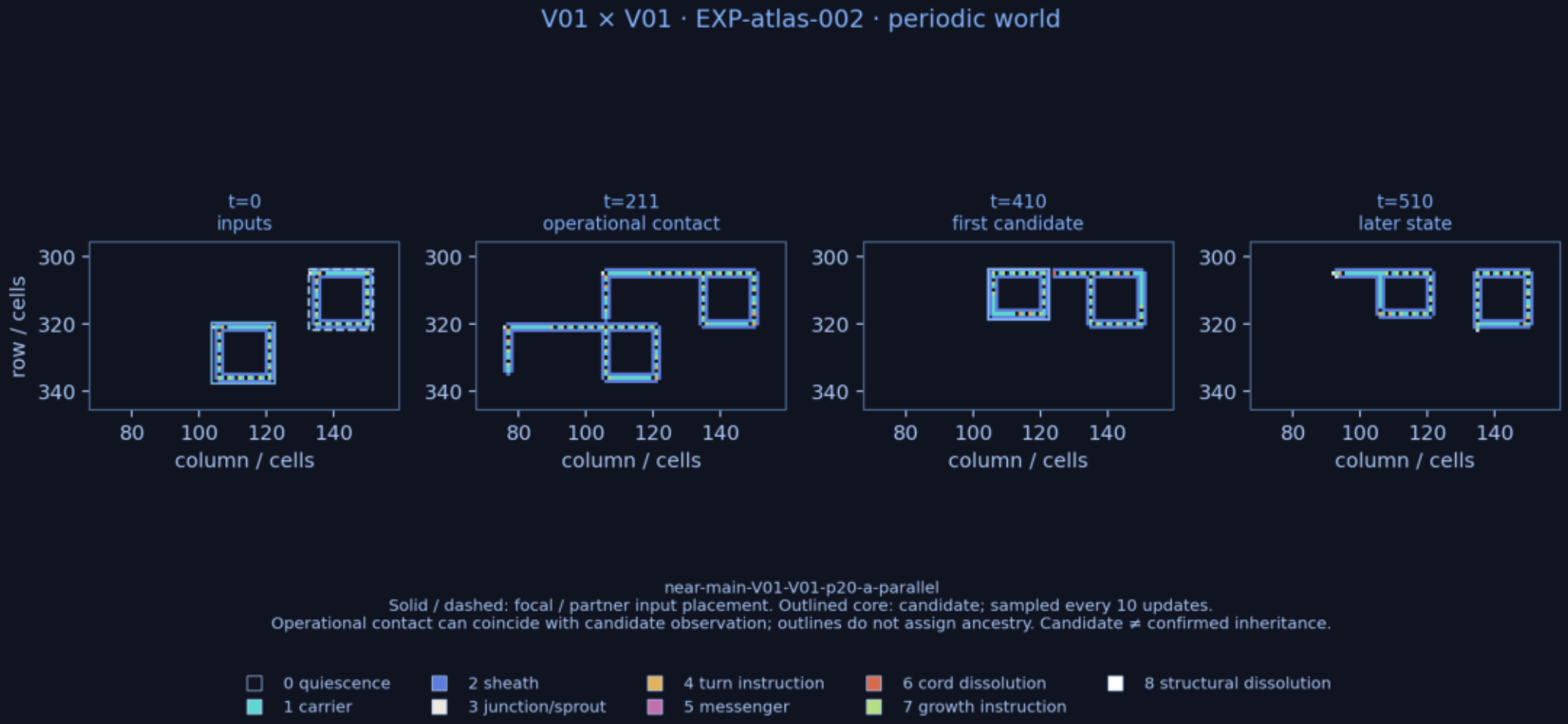


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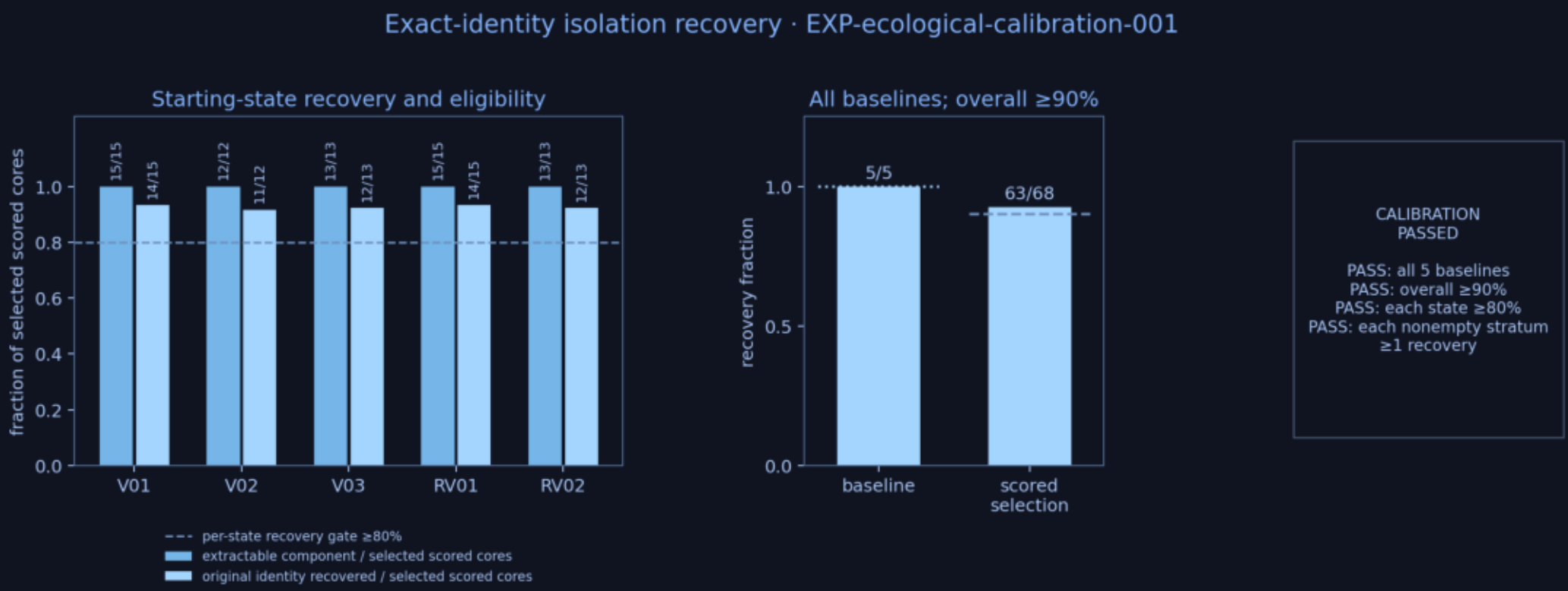


Figure: eco1-calibration-recovery. Source and regeneration: figures/manifest.json.

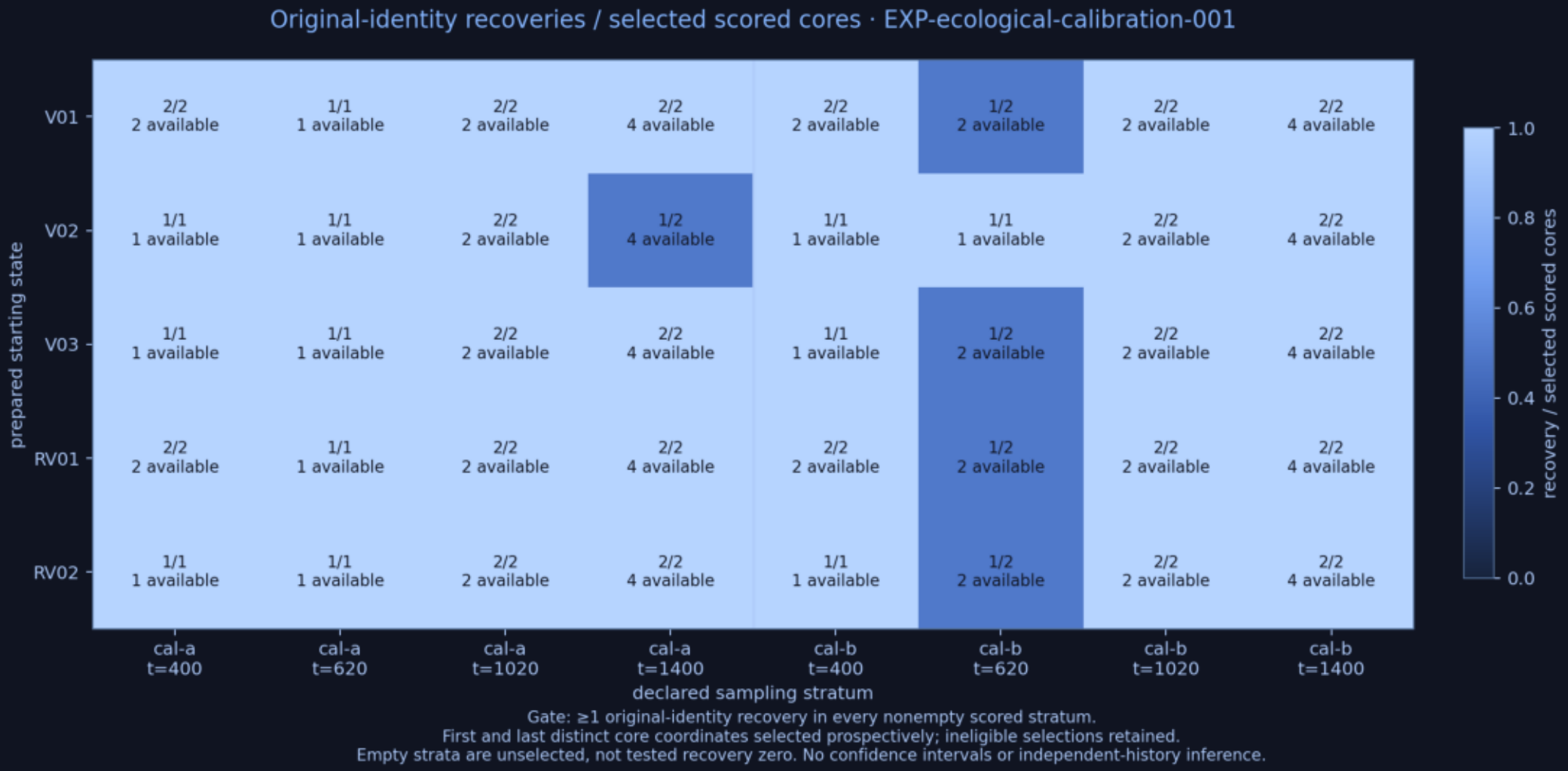
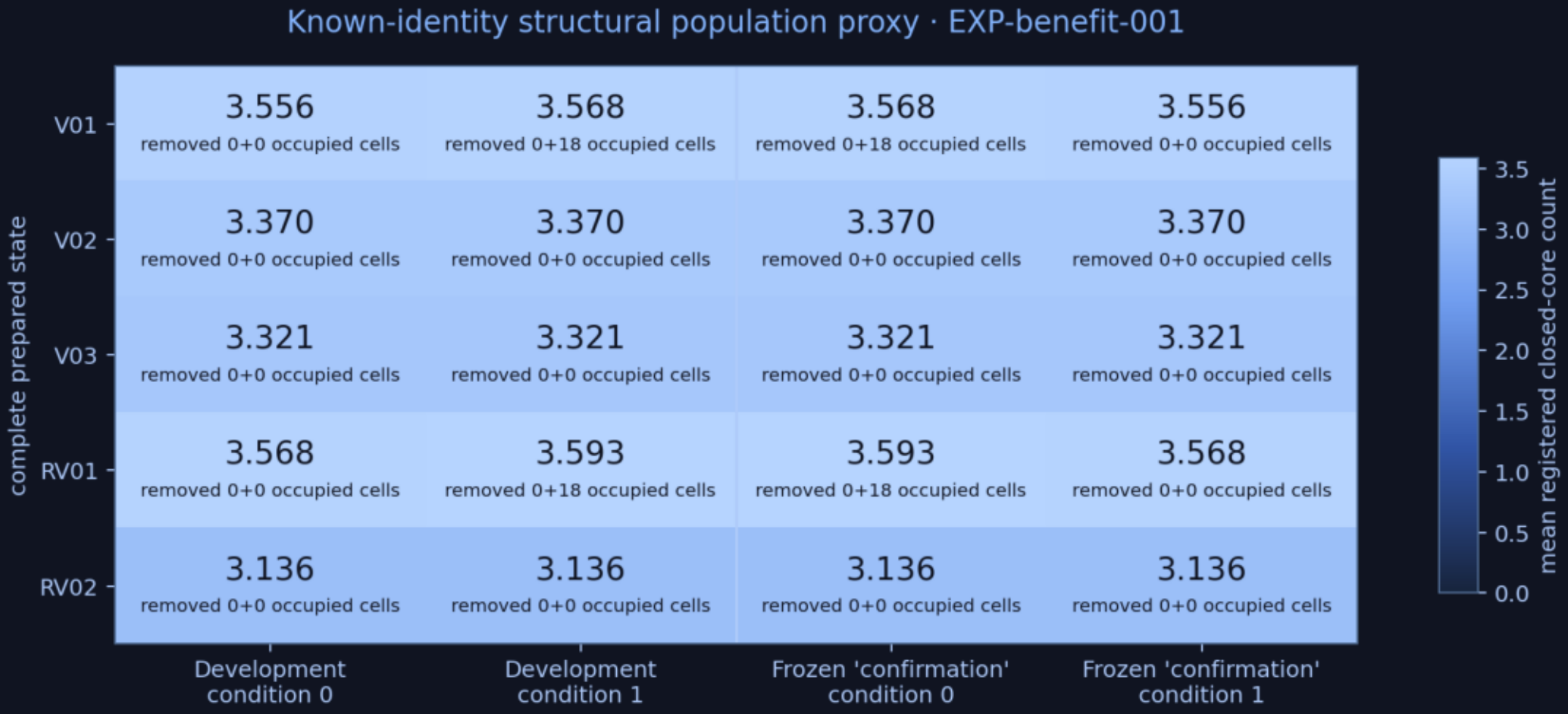


Figure: eco1-calibration-strata. Source and regeneration: figures/manifest.json.



Each mean uses 81 equally weighted samples (t=0,20,...,1600); occupied-cell removals shown at t=600+1000.
36/40 clearings removed zero occupied cells; 0/40 immediately changed registered core counts.
Development and confirmation panel means are identical.
Frozen 'confirmation' states are rotational symmetry duplicates; independent exposure confirmation remains unresolved.
Fixed four-identity registry; V01/RV01 share an identity. Other valid and unknown identities are excluded.
Deterministic conditions; no population replication, ancestry assignment, total viable-population or disturbance-resilience claim.

Figure: eco1-score-matrix. Source and regeneration: figures/manifest.json.

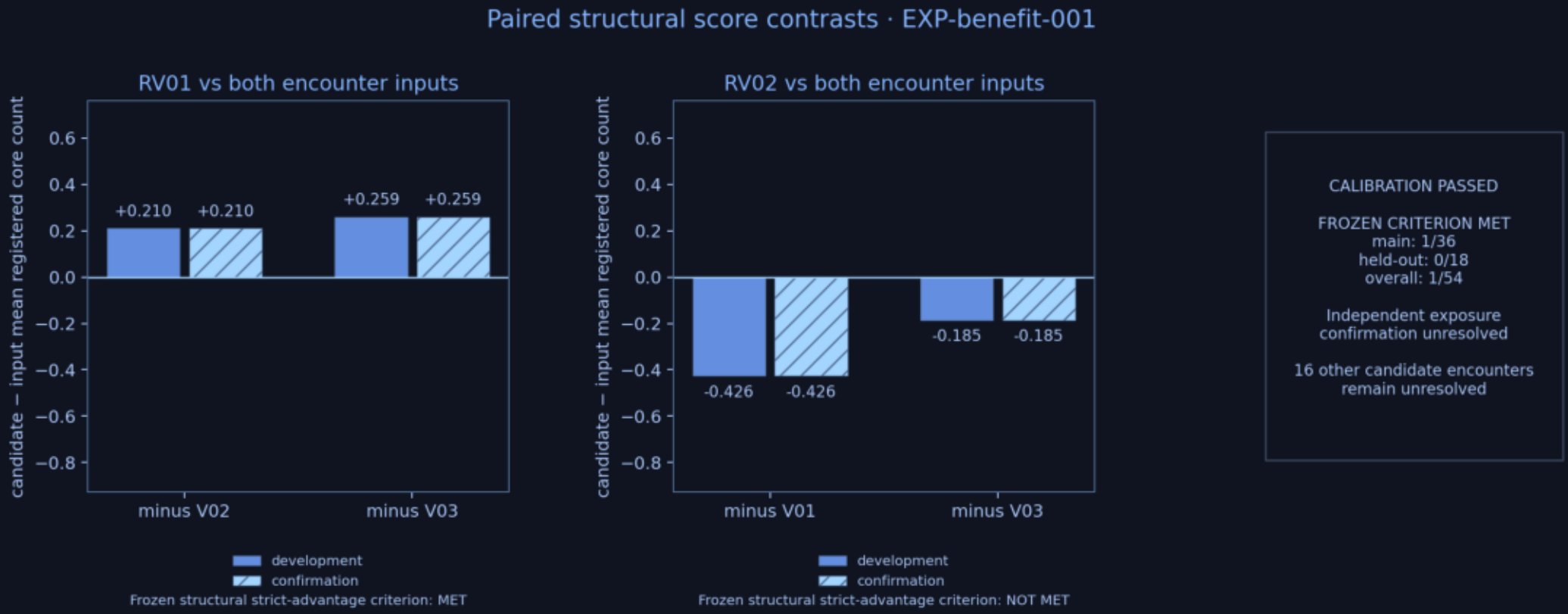


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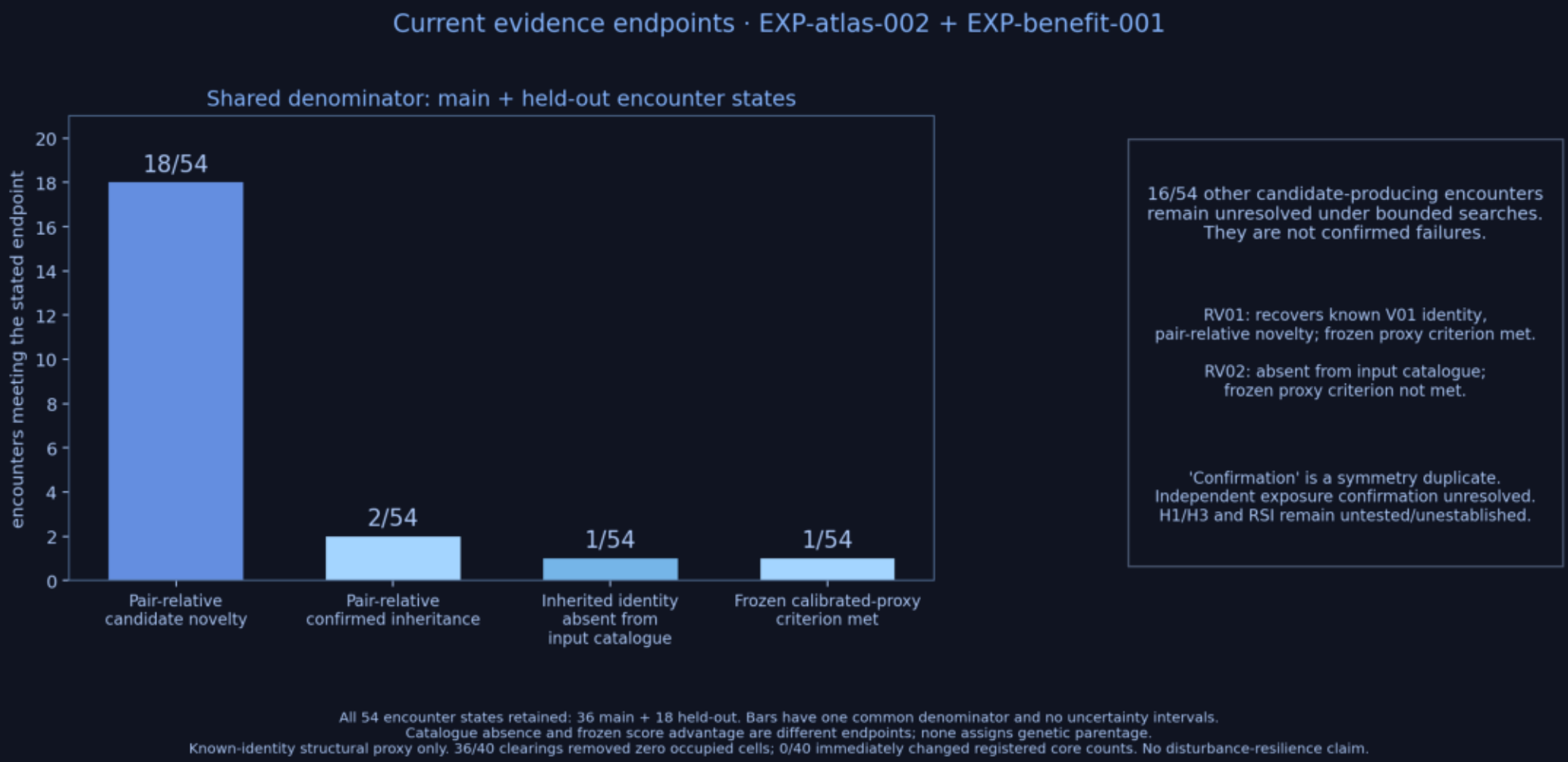


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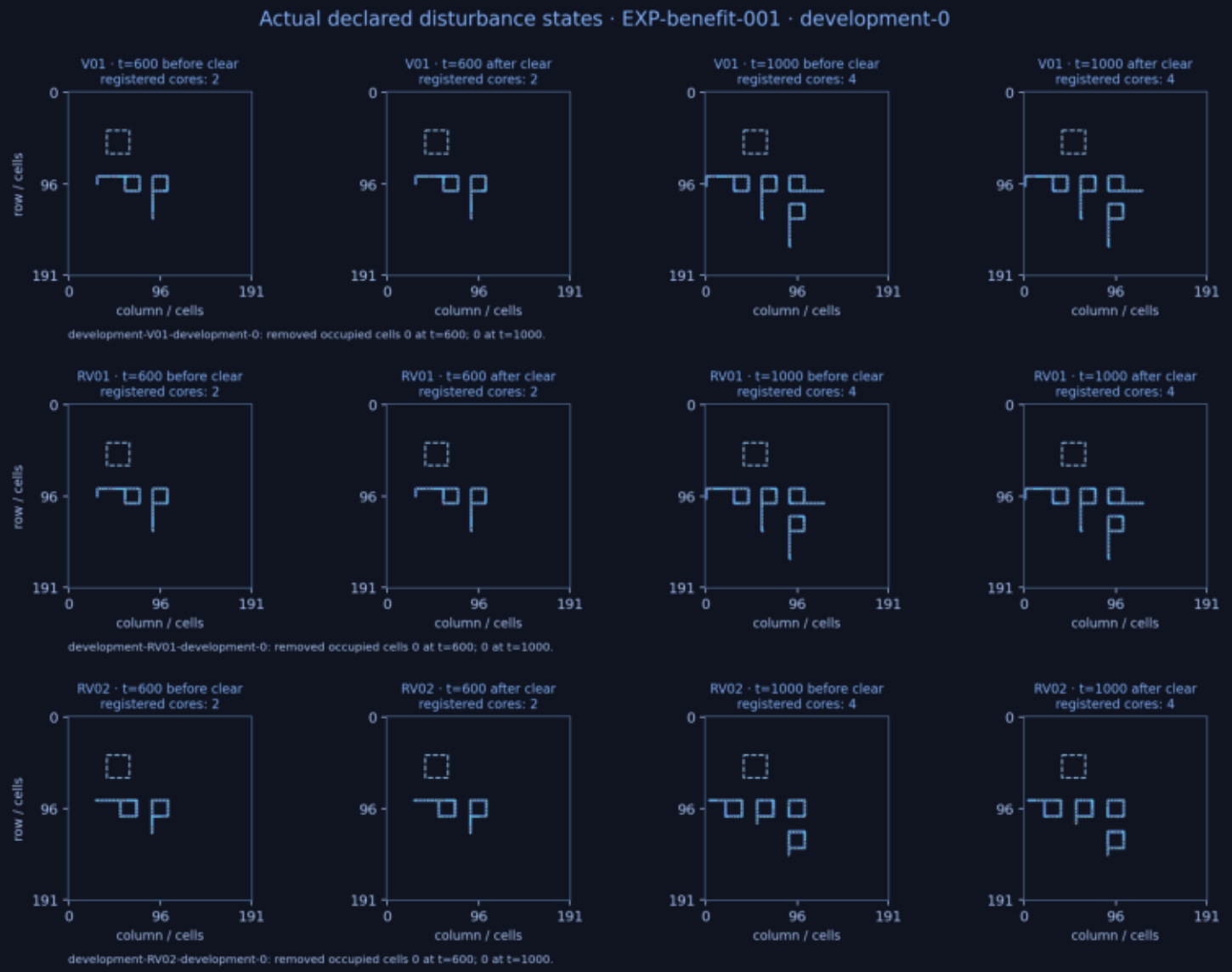


Figure: eco1-disturbance-states. Source and regeneration: figures/manifest.json.